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Dale et al.

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(54) **GAMING SYSTEM AND METHOD WITH A PERSISTENT ELEMENT FEATURE**

(71) Applicant: **LNW Gaming, Inc.**, Las Vegas, NV (US)

(72) Inventors: **Kellen Dale**, Reno, NV (US); **Amy Robards**, Sparks, NV (US); **Colbee V. Jones**, Reno, NV (US); **Dougal Austin**, Reno, NV (US)

(73) Assignee: **LNW Gaming, Inc.**, Las Vegas, NV (US)

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G07F 17/00 (2006.01)
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(52) **U.S. Cl.**
CPC ***G07F 17/3267*** (2013.01); ***G07F 17/3213*** (2013.01); ***G07F 17/3258*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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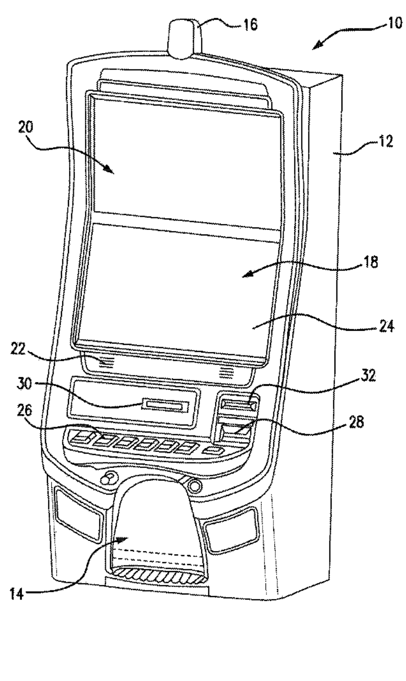
Primary Examiner — Paul A D'Agostino

(57) **ABSTRACT**

There are provided a gaming machine and method that utilize game-logic circuitry and a presentation assembly configured to present a plurality of symbol-bearing reels, an array, and a plurality of persistent elements. The plurality of reels are spun and stopped to land symbols from the reels in the array. In response to the landed symbols including at least one accumulation symbol, an animation of addition of the accumulation symbol to one of the plurality of persistent elements is presented. A random determination whether or not to trigger one or more game features is made. If the determination is to trigger the one or more game features, a bonus mode is entered and the one or more game features are implemented via the game-logic circuitry for subsequent spins of the game, for any free games awarded during the subsequent spins, or any combination thereof until the bonus mode is terminated.

18 Claims, 13 Drawing Sheets



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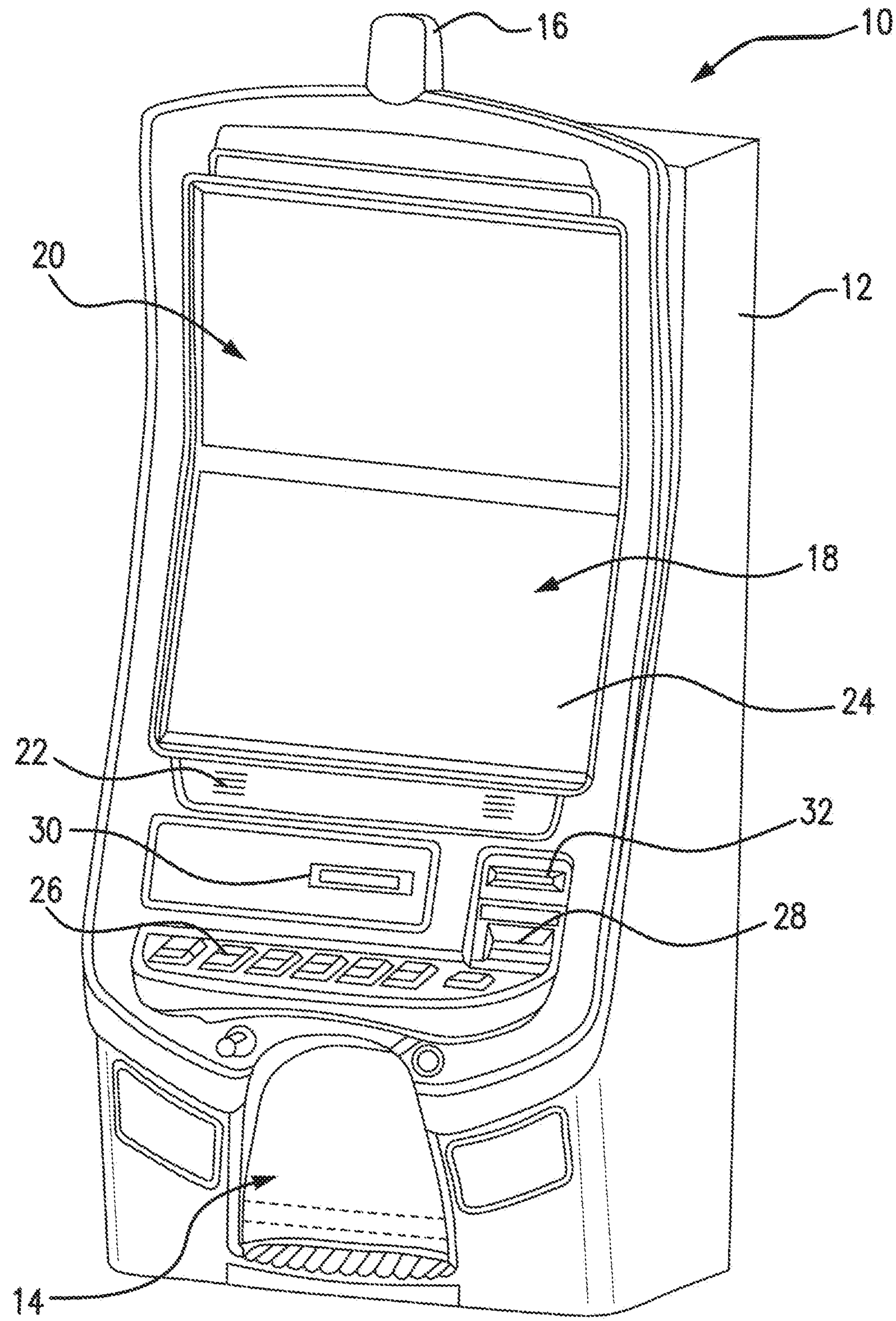

FIG. 1

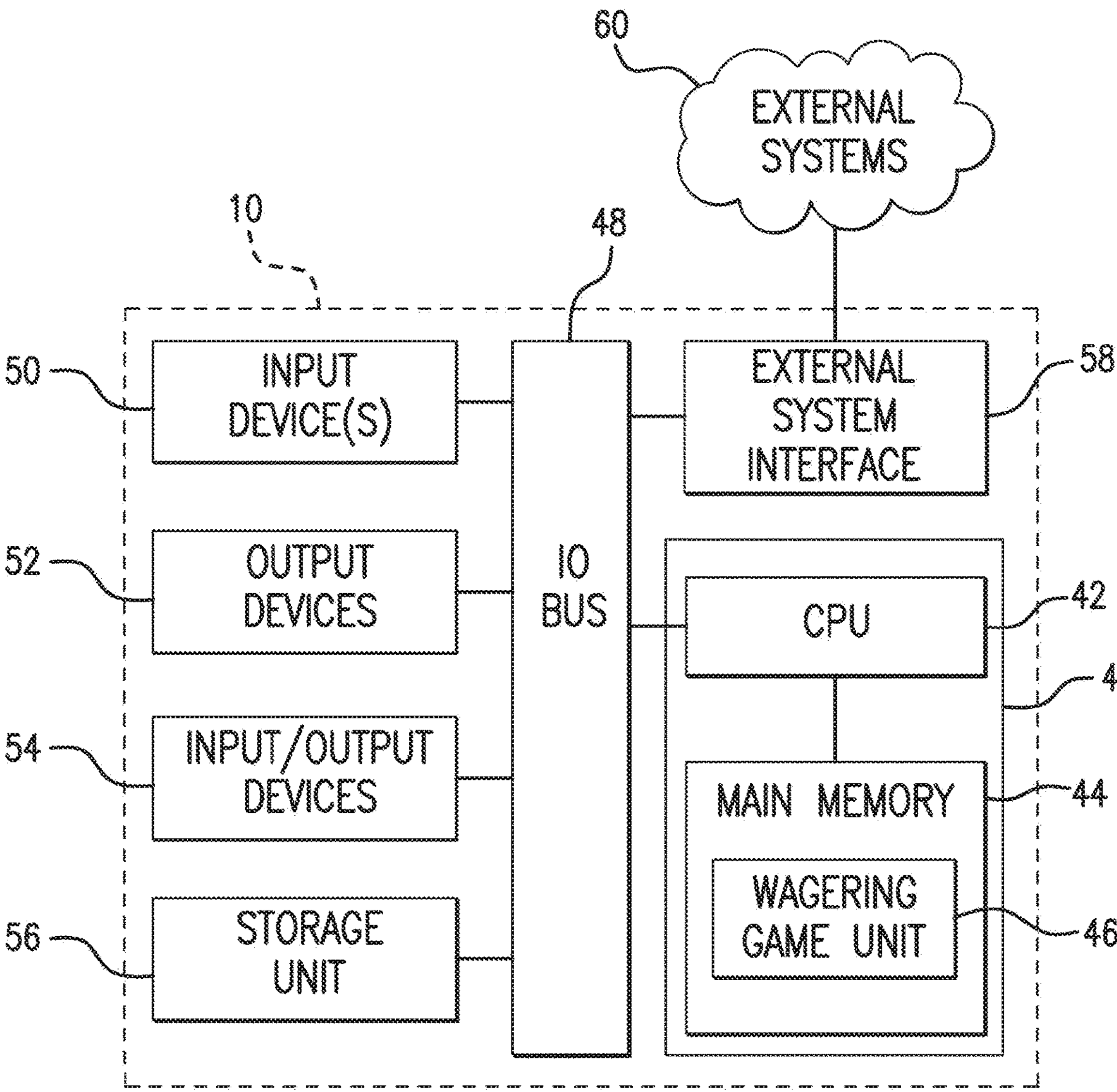
60
EXTERNAL SYSTEMS
10
48
50
INPUT DEVICE(S)
IO BUS
EXTERNAL SYSTEM INTERFACE
58
52
OUTPUT DEVICES
CPU
42
40
54
INPUT/OUTPUT DEVICES
MAIN MEMORY
44
56
STORAGE UNIT
WAGERING GAME UNIT
46

FIG. 2

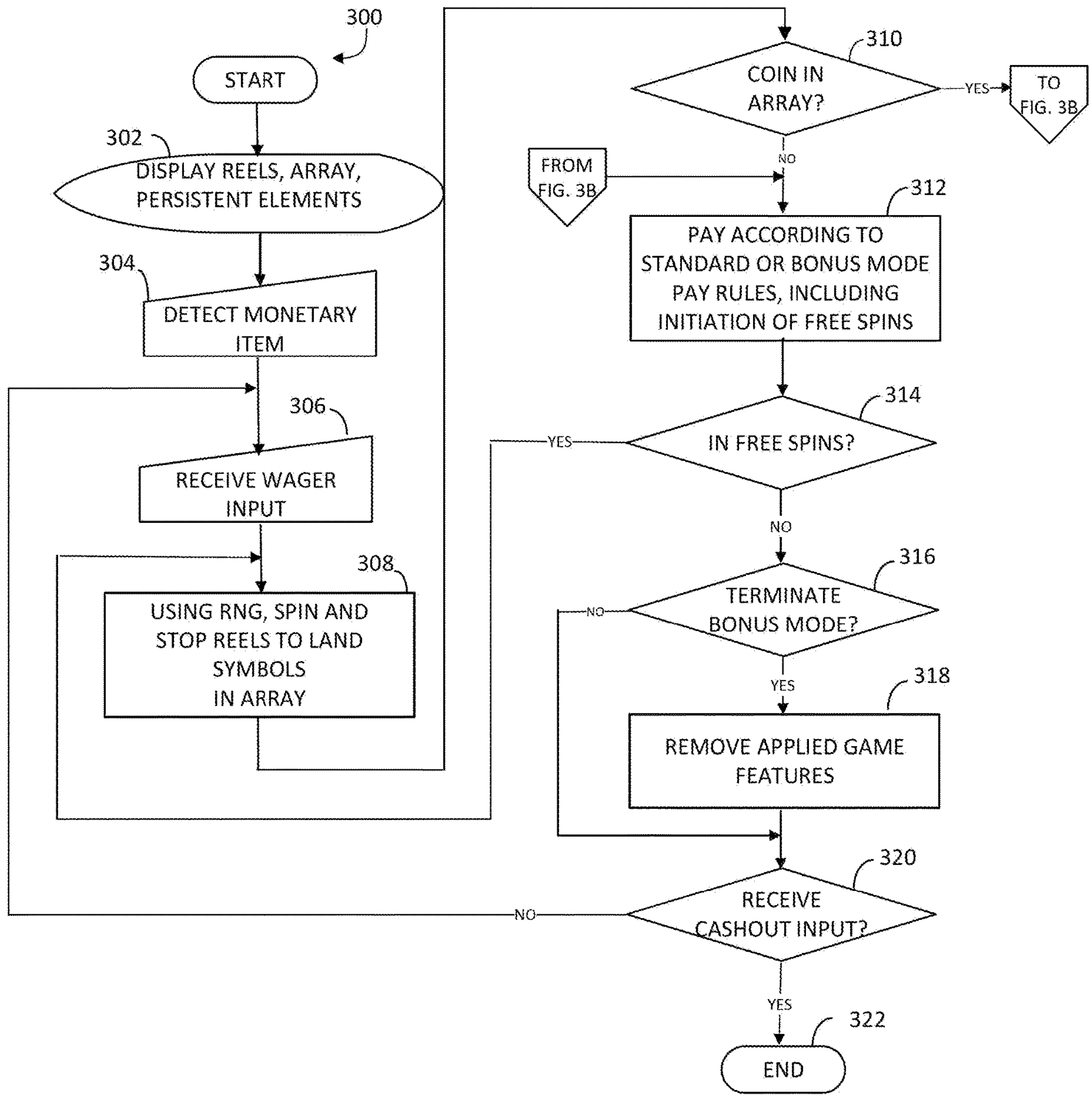
300
START
302
DISPLAY REELS, ARRAY, PERSISTENT ELEMENTS
304
DETECT MONETARY ITEM
306
RECEIVE WAGER INPUT
308
USING RNG, SPIN AND STOP REELS TO LAND SYMBOLS IN ARRAY
310
COIN IN ARRAY?
YES
TO FIG. 3B
NO
FROM FIG. 3B
312
PAY ACCORDING TO STANDARD OR BONUS MODE PAY RULES, INCLUDING INITIATION OF FREE SPINS
314
IN FREE SPINS?
YES
NO
316
TERMINATE BONUS MODE?
NO
YES
318
REMOVE APPLIED GAME FEATURES
320
RECEIVE CASHOUT INPUT?
NO
YES
322
END

FIG. 3A

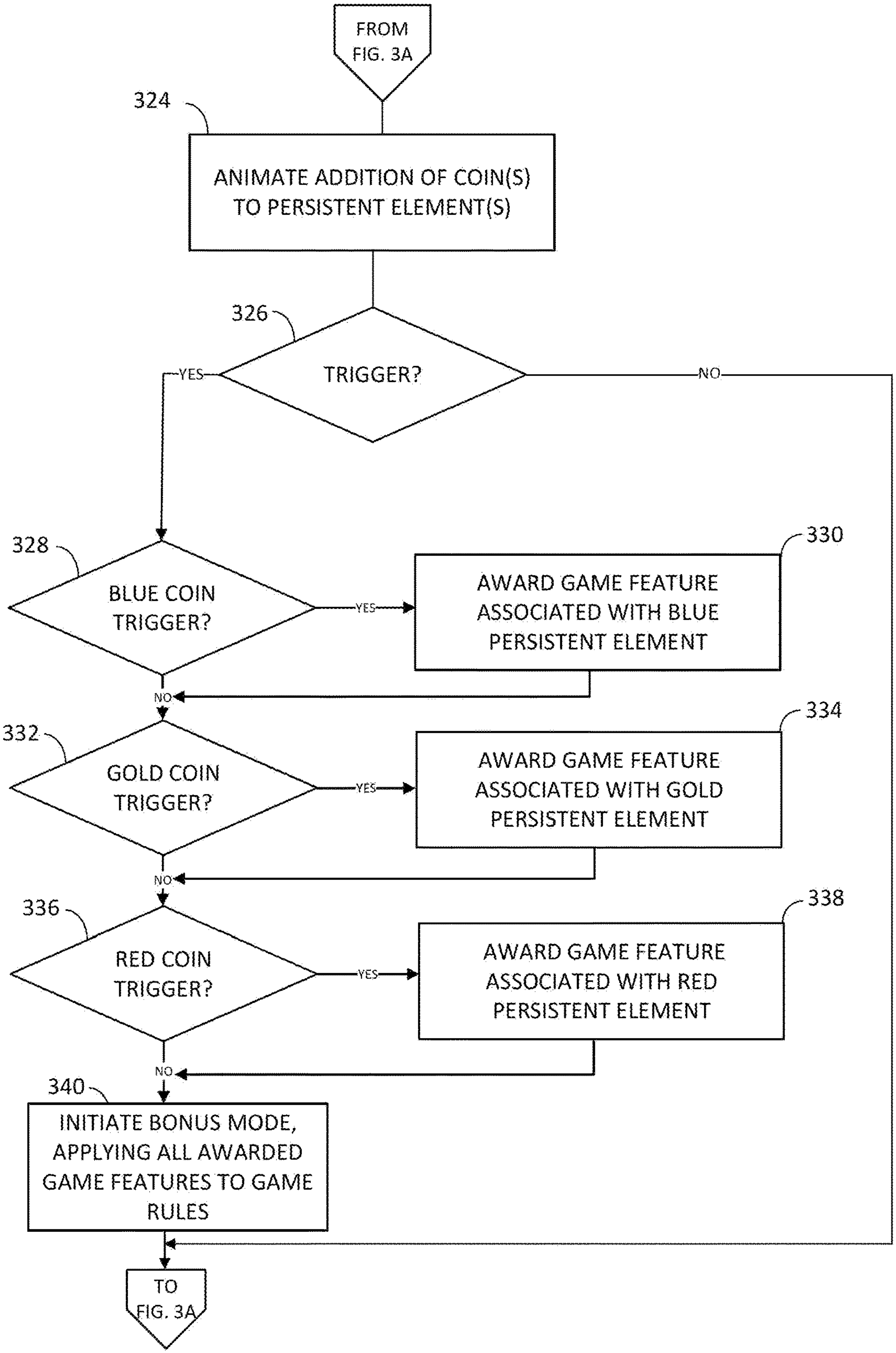
FROM FIG. 3A
324
ANIMATE ADDITION OF COIN(S) TO PERSISTENT ELEMENT(S)
326
TRIGGER?
YES
NO
328
BLUE COIN TRIGGER?
YES
330
AWARD GAME FEATURE ASSOCIATED WITH BLUE PERSISTENT ELEMENT
NO
332
GOLD COIN TRIGGER?
YES
334
AWARD GAME FEATURE ASSOCIATED WITH GOLD PERSISTENT ELEMENT
NO
336
RED COIN TRIGGER?
YES
338
AWARD GAME FEATURE ASSOCIATED WITH RED PERSISTENT ELEMENT
NO
340
INITIATE BONUS MODE, APPLYING ALL AWARDED GAME FEATURES TO GAME RULES
TO FIG. 3A

FIG. 3B

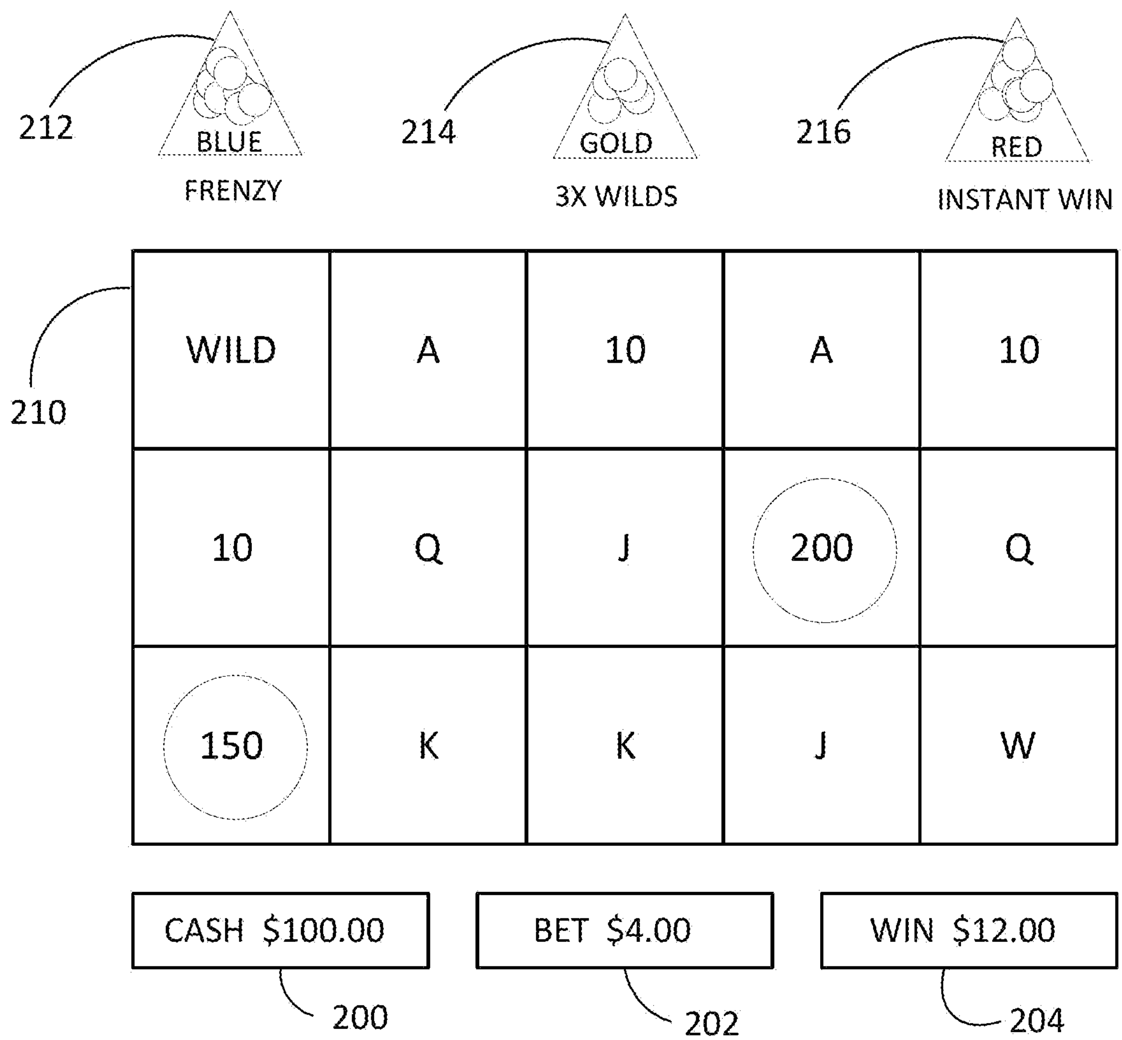
212
BLUE
FRENZY
214
GOLD
3X WILDS
216
RED
INSTANT WIN
210
WILD
A
10
A
10
10
Q
J
200
Q
150
K
K
J
W
CASH $100.00
BET $4.00
WIN $12.00
200
202
204

FIG. 4

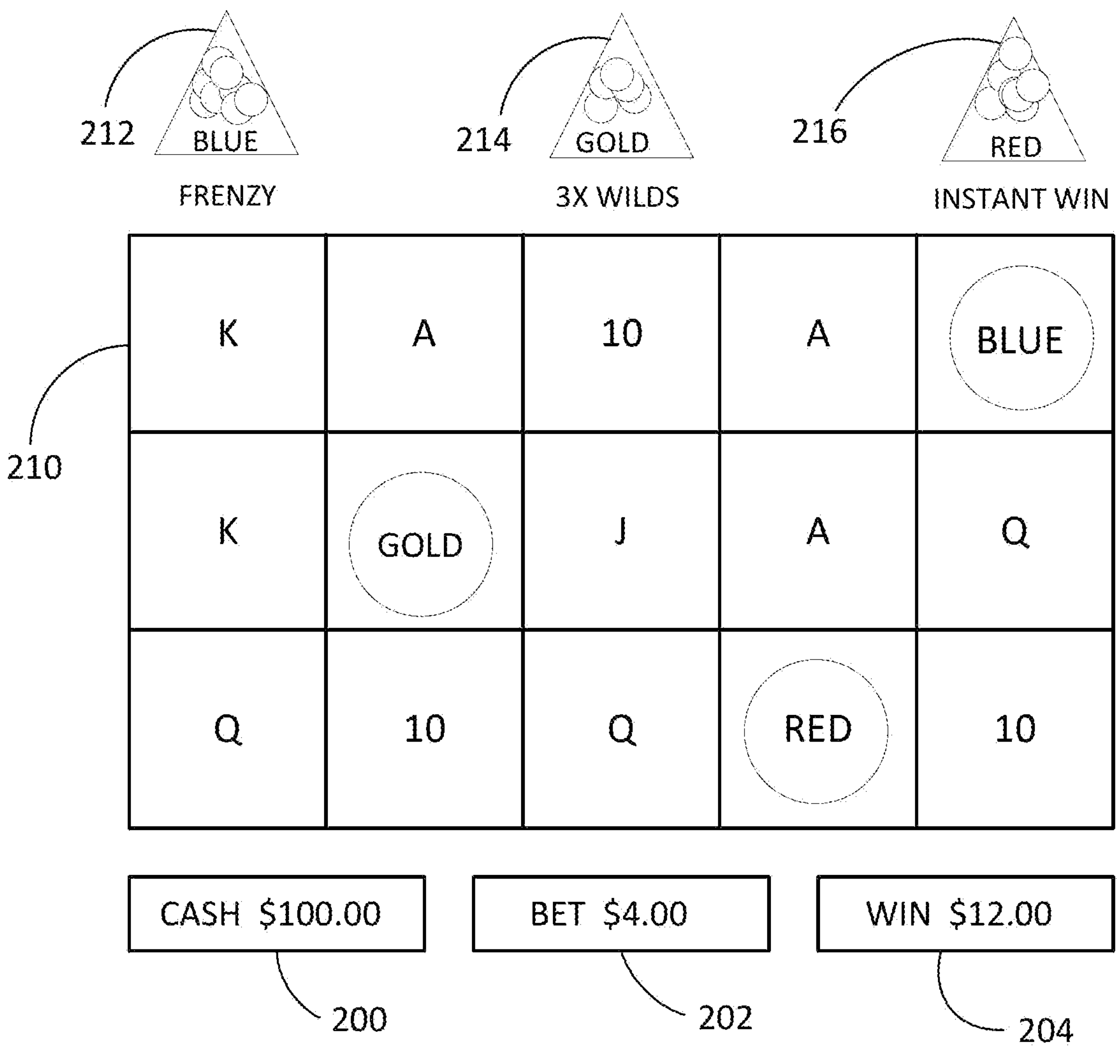
212
BLUE
FRENZY
214
GOLD
3X WILDS
216
RED
INSTANT WIN
210
K
A
10
A
BLUE
K
GOLD
J
A
Q
Q
10
Q
RED
10
CASH $100.00
BET $4.00
WIN $12.00
200
202
204

FIG. 5

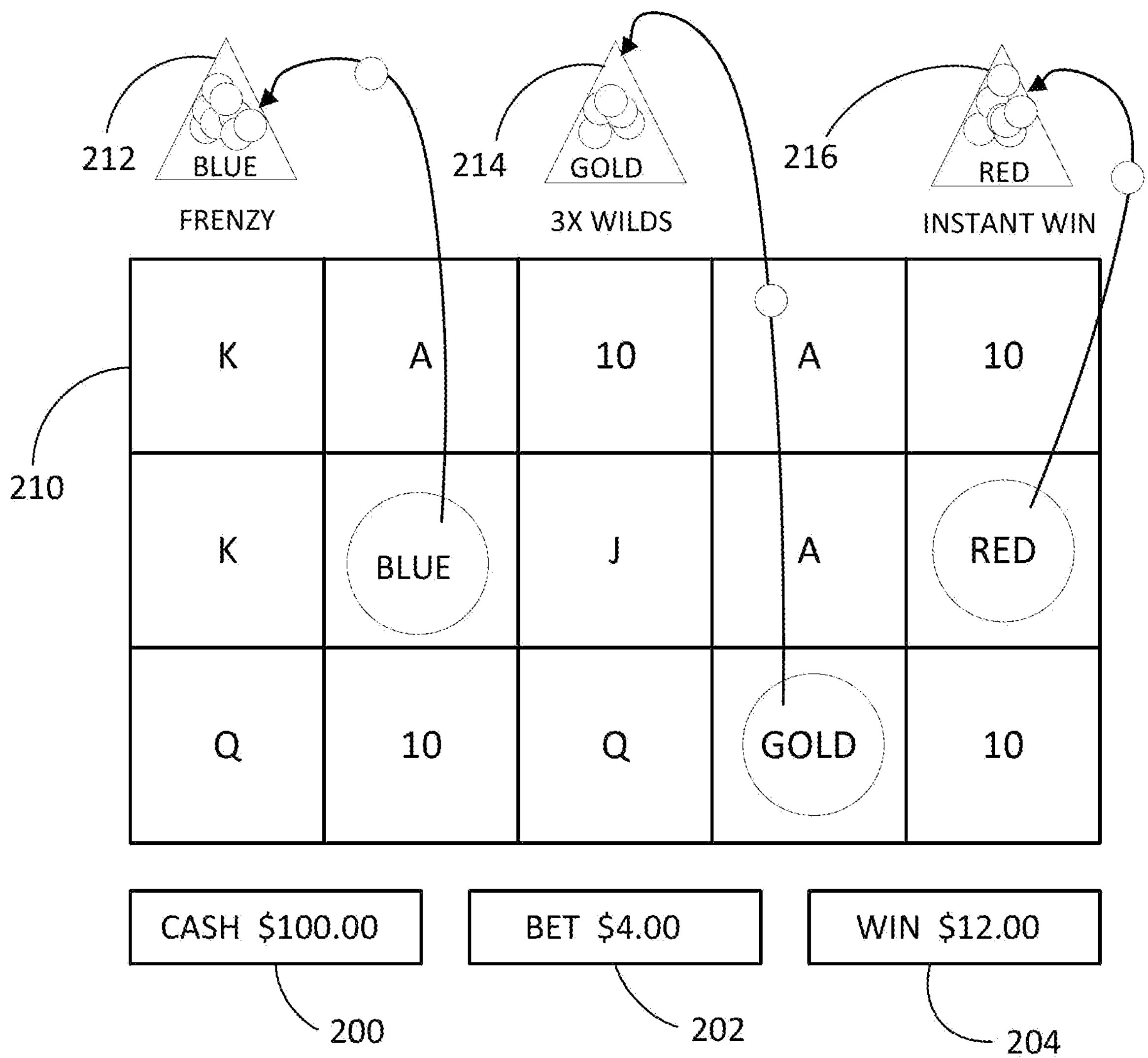
212
BLUE
FRENZY
214
GOLD
3X WILDS
216
RED
INSTANT WIN
210
K
A
10
A
10
K
BLUE
J
A
RED
Q
10
Q
GOLD
10
CASH $100.00
BET $4.00
WIN $12.00
200
202
204

FIG. 6

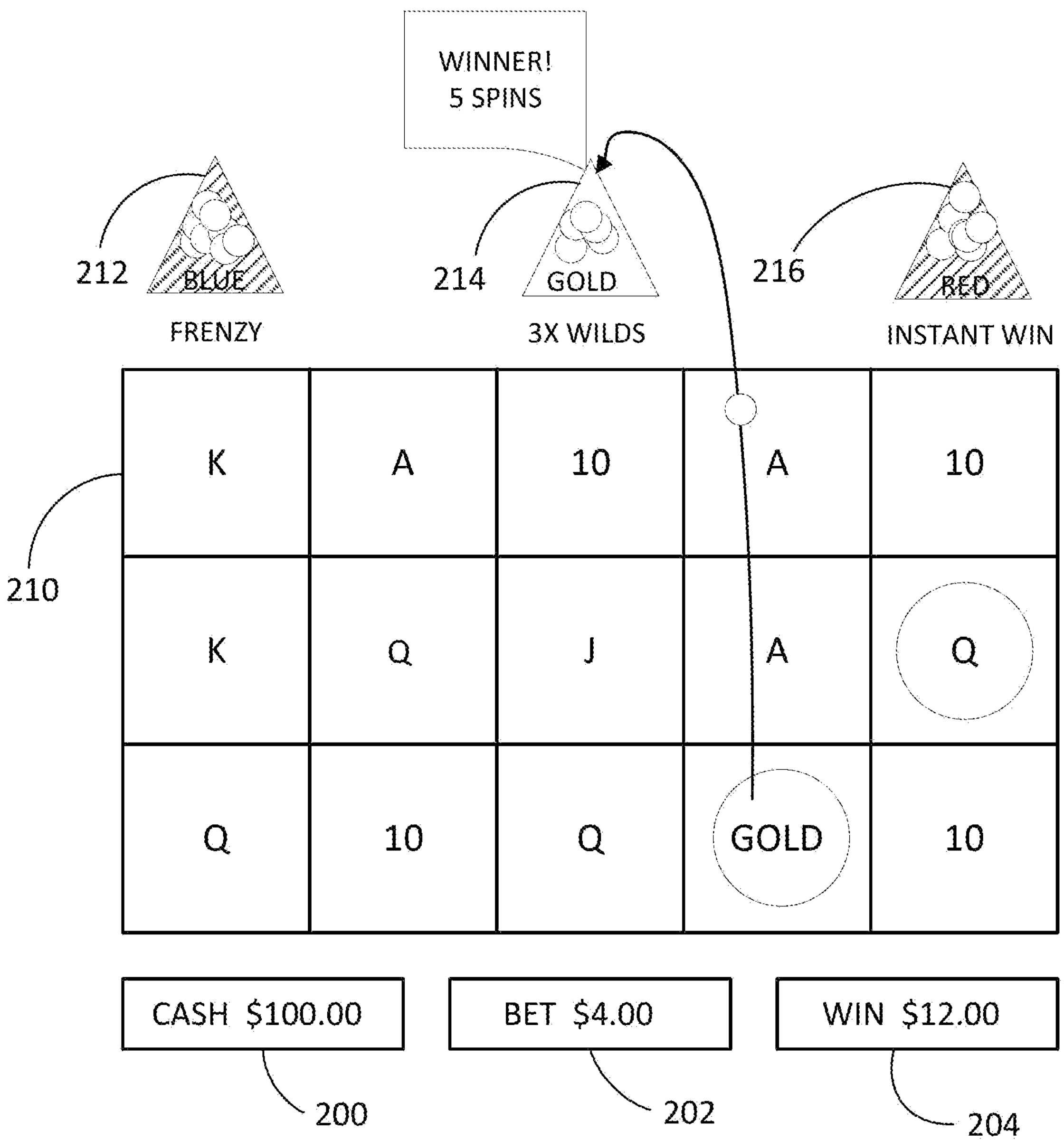
WINNER!
5 SPINS
212
BLUE
FRENZY
214
GOLD
3X WILDS
216
RED
INSTANT WIN
210
K
A
10
A
10
K
Q
J
A
Q
Q
10
Q
GOLD
10
CASH $100.00
BET $4.00
WIN $12.00
200
202
204

FIG. 7

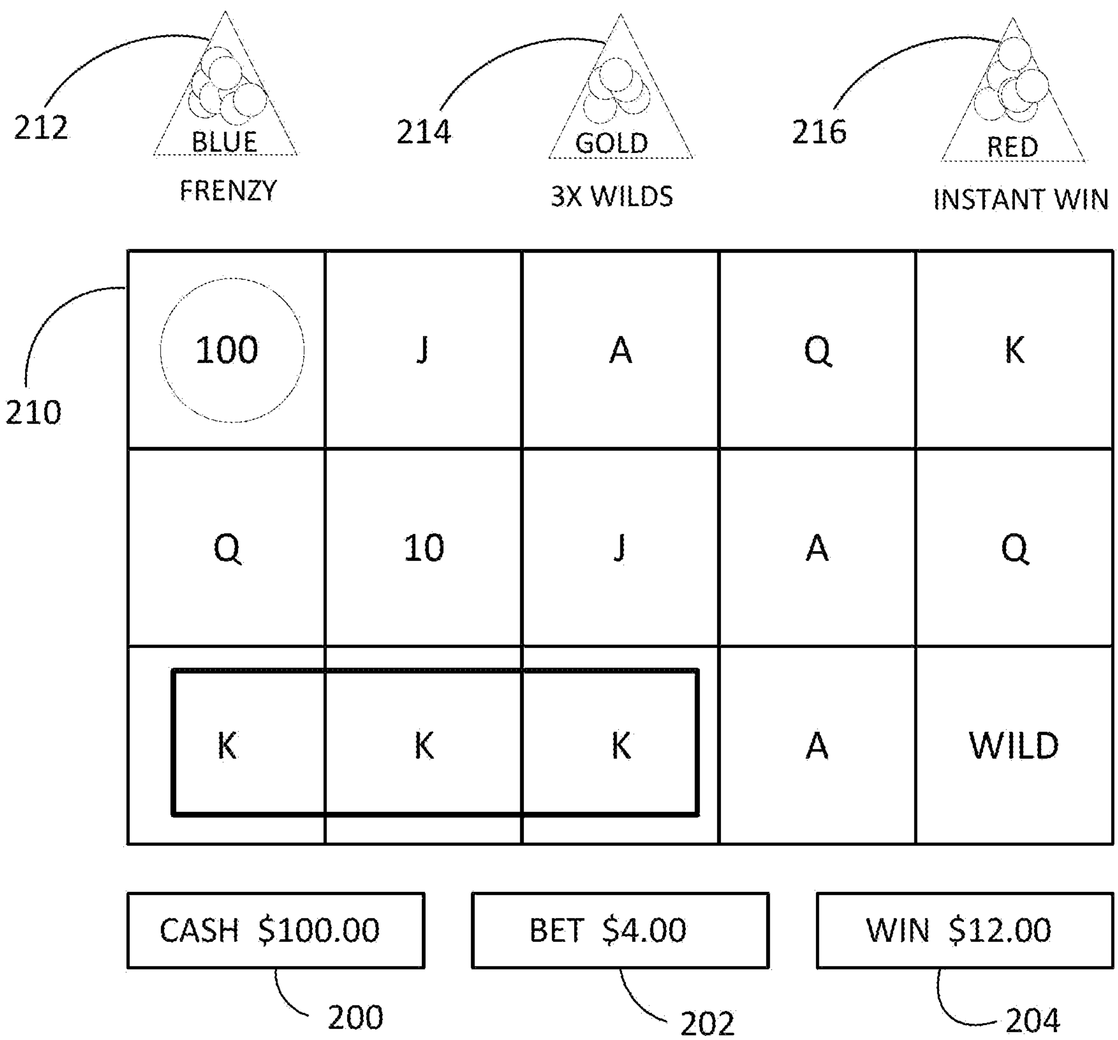
212
BLUE
FRENZY
214
GOLD
3X WILDS
216
RED
INSTANT WIN
210
100
J
A
Q
K
Q
10
J
A
Q
K
K
K
A
WILD
CASH $100.00
BET $4.00
WIN $12.00
200
202
204

FIG. 8

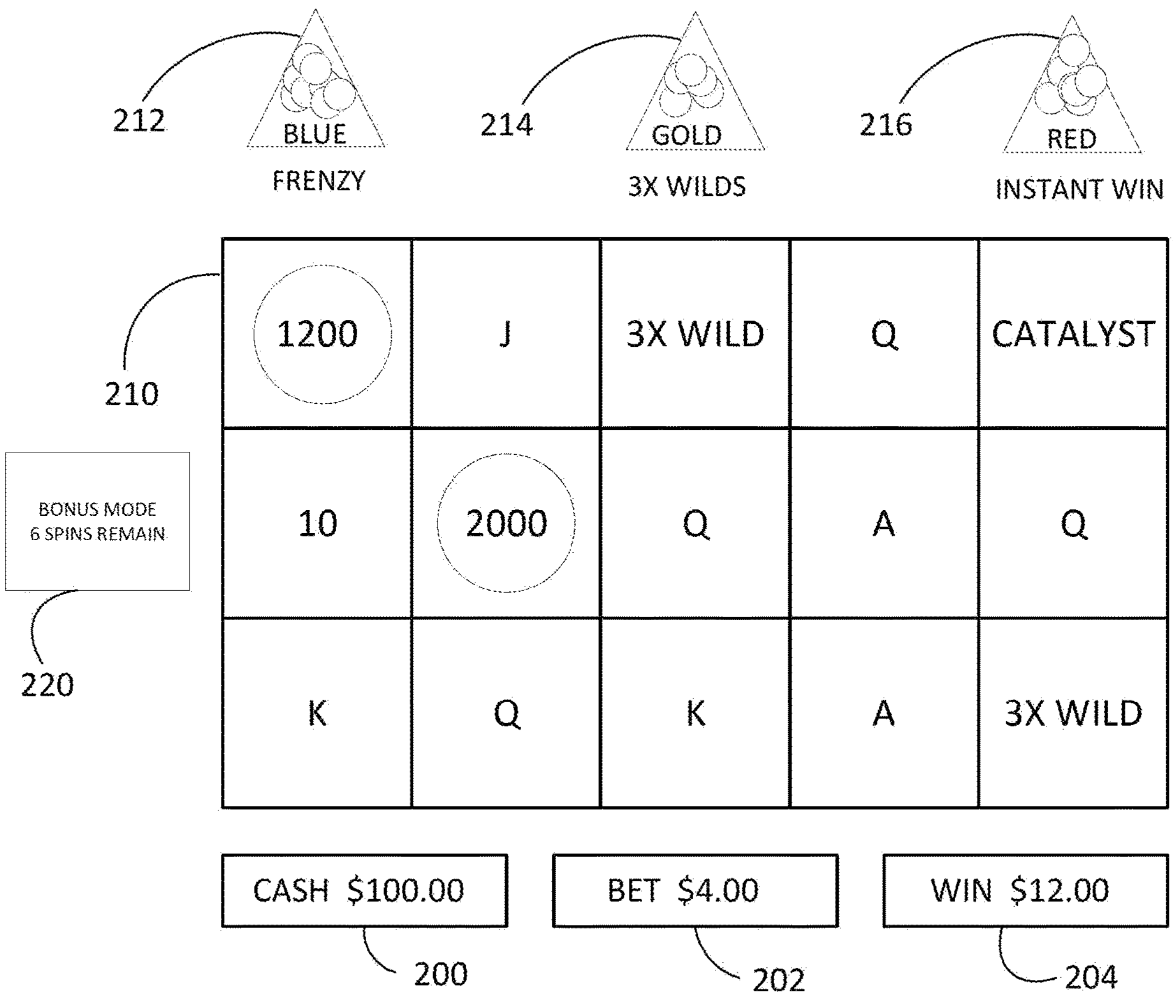
212
BLUE
FRENZY
214
GOLD
3X WILDS
216
RED
INSTANT WIN
210
1200
J
3X WILD
Q
CATALYST
BONUS MODE
6 SPINS REMAIN
10
2000
Q
A
Q
220
K
Q
K
A
3X WILD
CASH $100.00
BET $4.00
WIN $12.00
200
202
204

FIG. 9

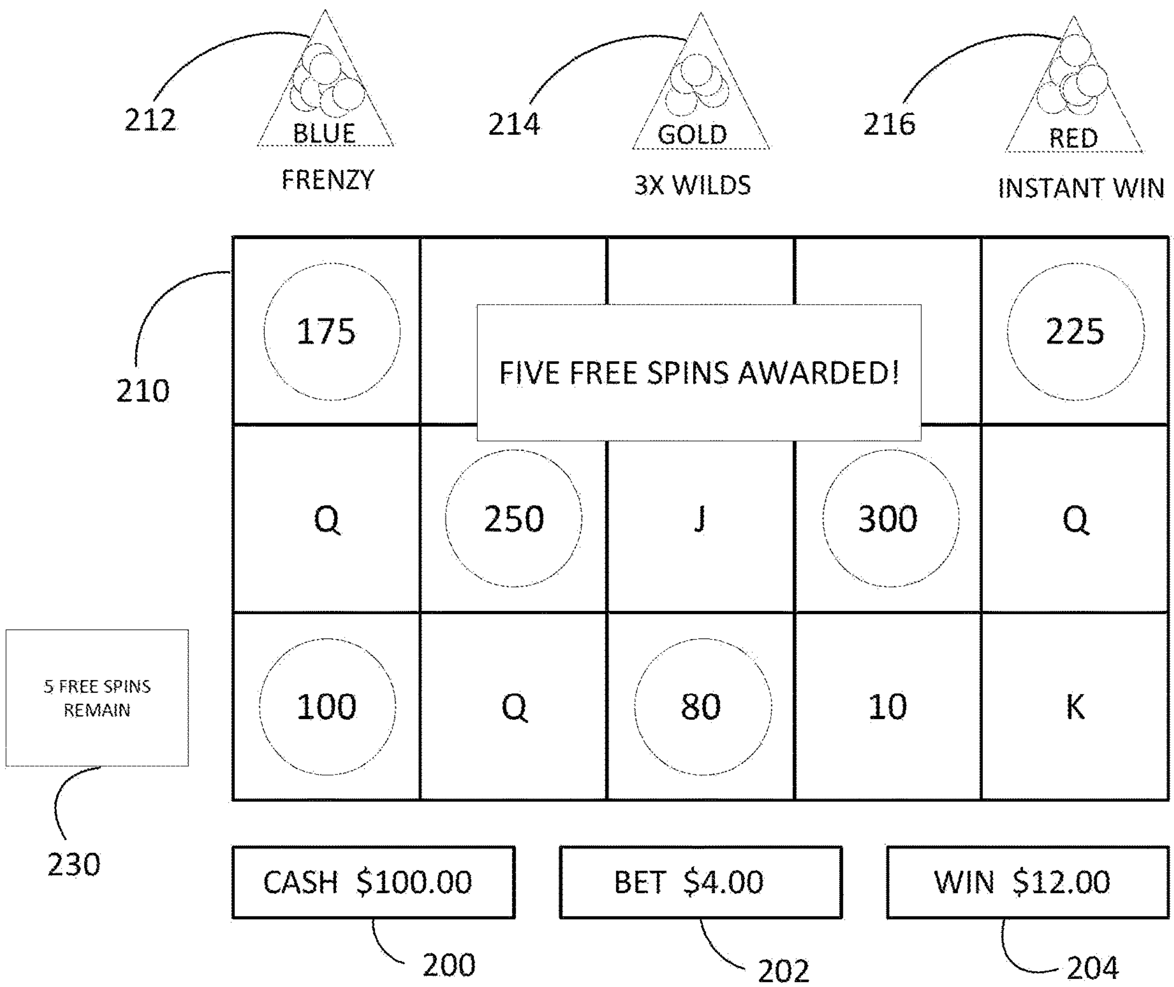
212
BLUE
FRENZY
214
GOLD
3X WILDS
216
RED
INSTANT WIN
210
175
225
FIVE FREE SPINS AWARDED!
Q
250
J
300
Q
100
Q
80
10
K
5 FREE SPINS REMAIN
230
CASH $100.00
BET $4.00
WIN $12.00
200
202
204

FIG. 10

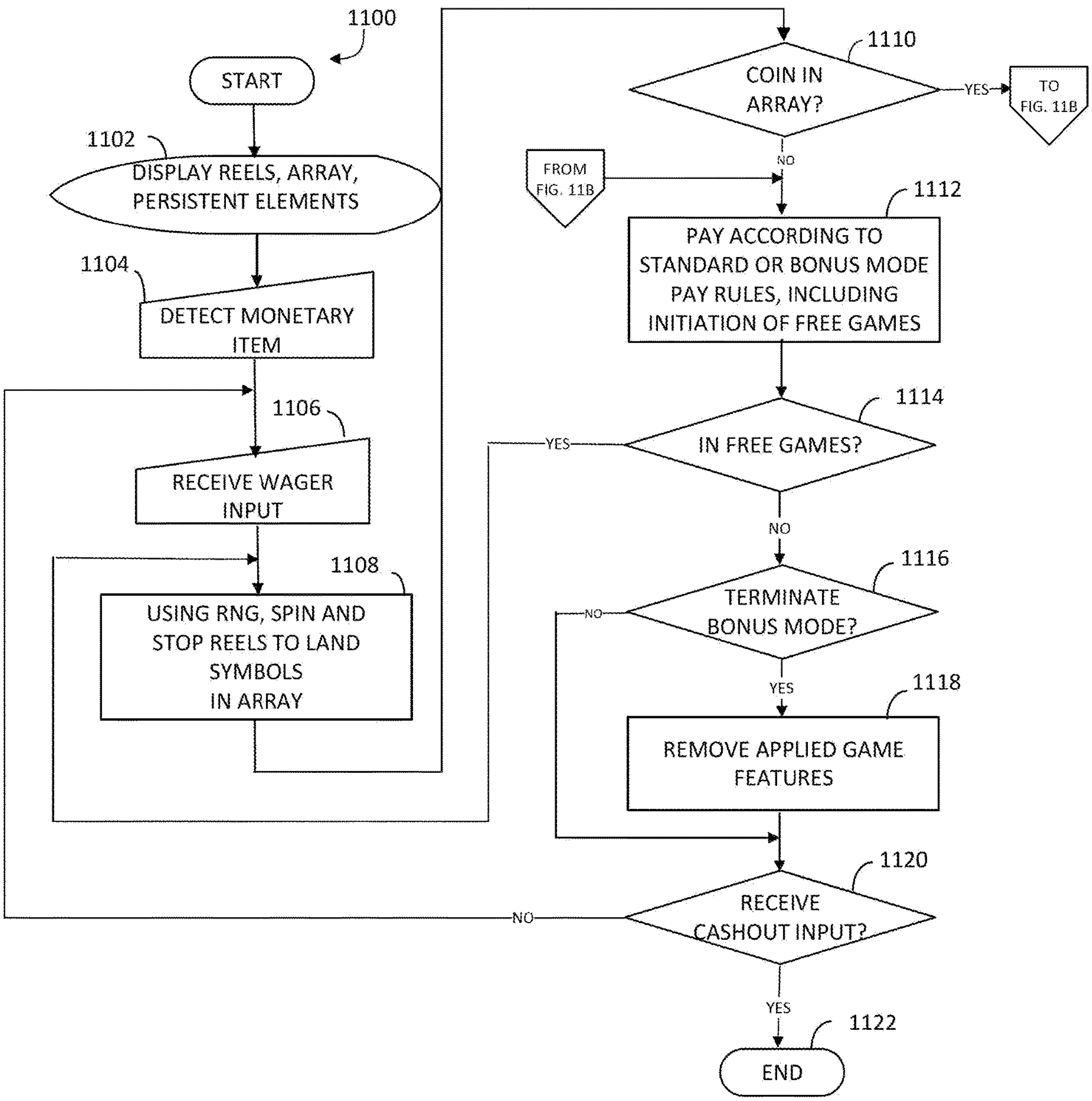
1100
START
1102
DISPLAY REELS, ARRAY, PERSISTENT ELEMENTS
1104
DETECT MONETARY ITEM
1106
RECEIVE WAGER INPUT
1108
USING RNG, SPIN AND STOP REELS TO LAND SYMBOLS IN ARRAY
1110
COIN IN ARRAY?
YES
TO FIG. 11B
NO
FROM FIG. 11B
1112
PAY ACCORDING TO STANDARD OR BONUS MODE PAY RULES, INCLUDING INITIATION OF FREE GAMES
1114
IN FREE GAMES?
YES
NO
1116
TERMINATE BONUS MODE?
NO
YES
1118
REMOVE APPLIED GAME FEATURES
1120
RECEIVE CASHOUT INPUT?
NO
YES
1122
END

FIG. 11A

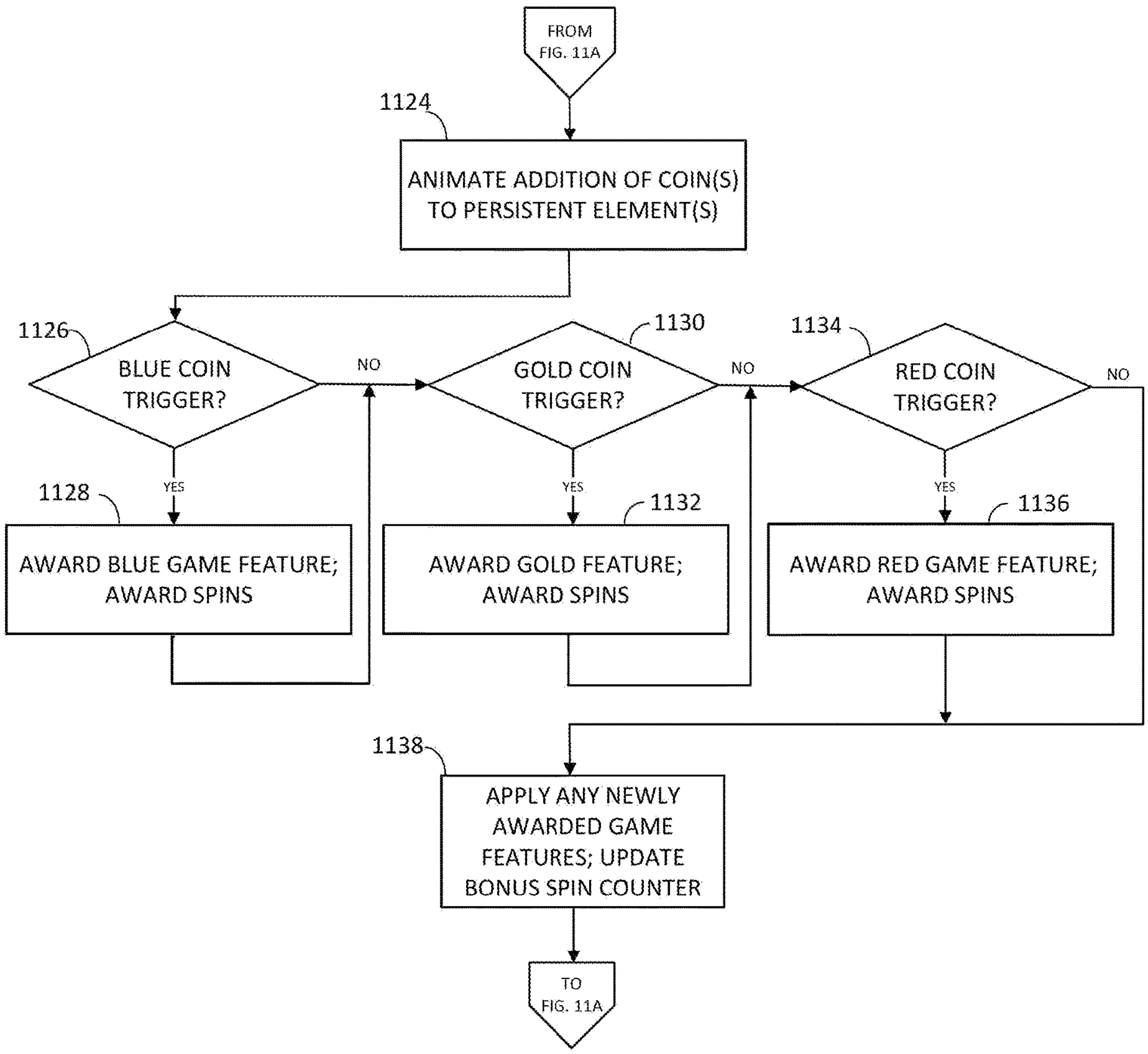
FROM
FIG. 11A
1124
ANIMATE ADDITION OF COIN(S) TO PERSISTENT ELEMENT(S)
1126
BLUE COIN TRIGGER?
NO
1130
GOLD COIN TRIGGER?
NO
1134
RED COIN TRIGGER?
NO
YES
1128
AWARD BLUE GAME FEATURE; AWARD SPINS
YES
1132
AWARD GOLD FEATURE; AWARD SPINS
YES
1136
AWARD RED GAME FEATURE; AWARD SPINS
1138
APPLY ANY NEWLY AWARDED GAME FEATURES; UPDATE BONUS SPIN COUNTER
TO
FIG. 11A

FIG. 11B

GAMING SYSTEM AND METHOD WITH A PERSISTENT ELEMENT FEATURE

RELATED APPLICATIONS

This application is related to U.S. patent application Ser. No. 18/153,644, entitled "Gaming System and Method With a Persistent Element Feature," filed on Jan. 12, 2023, which is hereby incorporated by reference in its entirety.

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FIELD OF THE INVENTION

The present invention relates to a technological improvement to gaming systems, gaming machines, and methods and, more particularly, to technological improvements in connection with a persistent element feature.

BACKGROUND OF THE INVENTION

The gaming industry depends upon player participation. Players are generally "hopeful" players who either think they are lucky or at least think they can get lucky—for a relatively small investment to play a game, they can get a disproportionately large return. To create this feeling of luck, a gaming apparatus relies upon an internal or external random element generator to generate one or more random elements such as random numbers. The gaming apparatus determines a game outcome based, at least in part, on the one or more random elements.

A significant technical challenge is to improve the operation of gaming apparatus and games played thereon, including the manner in which they leverage the underlying random element generator, by making them yield a negative return on investment in the long run (via a high quantity and/or frequency of player/apparatus interactions) and yet random and volatile enough to make players feel they can get lucky and win in the short run. Striking the right balance between yield versus randomness and volatility to create a feeling of luck involves addressing many technical problems, some of which can be at odds with one another. This luck factor is what appeals to core players and encourages prolonged and frequent player participation. As the industry matures, the creativity and ingenuity required to improve such operation of gaming apparatus and games grows accordingly.

Another significant technical challenge is to improve the operation of gaming apparatus and games played thereon by increasing processing speed and efficiency of usage of processing and/or memory resources. To make games more entertaining and exciting, they often offer the complexities of advanced graphics and special effects, multiple game features with different game formats, and multiple random outcome determinations per feature. The game formats may, for example, include picking games, reel spins, wheel spins, and other arcade-style play mechanics. Inefficiencies in processor execution of the game software can slow down play of the game and prevent a player from playing the game at their desired pace.

Yet another significant technical challenge is to provide a new and improved level of game play that uses new and improved gaming apparatus animations. Improved animations represent improvements to the underlying technology or technical field of gaming apparatus and, at the same time, have the effect of encouraging prolonged and frequent player participation.

SUMMARY OF THE INVENTION

There are provided a gaming machine and method that utilize game-logic circuitry and a presentation assembly configured to present a plurality of symbol-bearing reels, an array, and a plurality of persistent elements. The plurality of reels are spun and stopped to land symbols from the reels in the array. In response to the landed symbols including at least one accumulation symbol, an animation of addition of the accumulation symbol to one of the plurality of persistent elements is presented. A random determination whether or not to trigger one or more game features is made. If the determination is to trigger the one or more game features, a bonus mode is entered and the one or more game features are implemented via the game-logic circuitry for subsequent spins of the game, for any free games awarded during the subsequent spins, or any combination thereof until the bonus mode is terminated.

Additional aspects of the invention will be apparent to those of ordinary skill in the art in view of the detailed description of various embodiments, which is made with reference to the drawings, a brief description of which is provided below.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a front view of a free-standing gaming machine according to an embodiment of the present invention.

FIG. **2** is a block diagram of a gaming system according to an embodiment of the present invention.

FIGS. **3**A and **3**B are a flow diagram for a data processing method that corresponds to instructions executed by a controller, according to an embodiment of the present invention. FIG. **3**A relates to a base-game portion of a wagering game; FIG. **3**B relates to the awarding of one or more game features and a bonus mode that may be triggered as a result of the awarding of the game features.

FIGS. **4-10** are exemplary presentations of game spin outcomes resulting from the flow diagram in FIGS. **3**A-**3**B and FIGS. **11**A-**11**B.

FIGS. **11**A and **11**B are a flow diagram for a data processing method that corresponds to instructions executed by a controller, according to an alternate embodiment of the present invention. FIG. **11**A relates to a base-game portion of a wagering game; FIG. **11**B relates to the awarding of one or more game features and a bonus mode that may be triggered as a result of the awarding of the game features.

While the invention is susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and will be described in detail herein. It should be understood, however, that the invention is not intended to be limited to the particular forms disclosed. Rather, the invention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION

While this invention is susceptible of embodiment in many different forms, there is shown in the drawings and will herein be described in detail preferred embodiments of the invention with the understanding that the present disclosure is to be considered as an exemplification of the principles of the invention and is not intended to limit the broad aspect of the invention to the embodiments illustrated. For purposes of the present detailed description, the singular includes the plural and vice versa (unless specifically disclaimed); the words "and" and "or" shall be both conjunctive and disjunctive; the word "all" means "any and all"; the word "any" means "any and all"; and the word "including" means "including without limitation."

For purposes of the present detailed description, the terms "wagering game," "casino wagering game," "gambling," "slot game," "casino game," and the like include games in which a player places at risk a sum of money or other representation of value, whether or not redeemable for cash, on an event with an uncertain outcome, including without limitation those having some element of skill. In some embodiments, the wagering game involves wagers of real money, as found with typical land-based or online casino games. In other embodiments, the wagering game additionally, or alternatively, involves wagers of non-cash values, such as virtual currency, and therefore may be considered a social or casual game, such as would be typically available on a social networking web site, other web sites, across computer networks, or applications on mobile devices (e.g., phones, tablets, etc.). When provided in a social or casual game format, the wagering game may closely resemble a traditional casino game, or it may take another form that more closely resembles other types of social/casual games.

Referring to FIG. **1**, there is shown a gaming machine **10** similar to those operated in gaming establishments, such as casinos. With regard to the present invention, the gaming machine **10** may be any type of gaming terminal or machine and may have varying structures and methods of operation. For example, in some aspects, the gaming machine **10** is an electromechanical gaming terminal configured to play mechanical slots, whereas in other aspects, the gaming machine is an electronic gaming terminal configured to play a video casino game, such as slots, keno, poker, blackjack, roulette, craps, etc. The gaming machine **10** may take any suitable form, such as floor-standing models as shown, handheld mobile units, bartop models, workstation-type console models, etc. Further, the gaming machine **10** may be primarily dedicated for use in playing wagering games, or may include non-dedicated devices, such as mobile phones, personal digital assistants, personal computers, etc. Exemplary types of gaming machines are disclosed in U.S. Pat. Nos. 6,517,433, 8,057,303, and 8,226,459, which are incorporated herein by reference in their entireties.

The gaming machine **10** illustrated in FIG. **1** comprises a gaming cabinet **12** that securely houses various input devices, output devices, input/output devices, internal electronic/electromechanical components, and wiring. The cabinet **12** includes exterior walls, interior walls, and shelves for mounting the internal components and managing the wiring, and one or more front doors that are locked and require a physical or electronic key to gain access to the interior compartment of the cabinet **12** behind the locked door. The cabinet **12** forms an alcove **14** configured to store one or more beverages or personal items of a player. A notification mechanism **16**, such as a candle or tower light, is mounted to the top of the cabinet **12**. It flashes to alert an attendant that change is needed, a hand pay is requested, or there is a potential problem with the gaming machine **10**.

The input devices, output devices, and input/output devices are disposed on, and securely coupled to, the cabinet **12**. By way of example, the output devices include a primary presentation device **18**, a secondary presentation device **20**, and one or more audio speakers **22**. The primary presentation device **18** or the secondary presentation device **20** may be a mechanical-reel display device, a video display device, or a combination thereof. In one such combination disclosed in U.S. Pat. No. 6,517,433, a transmissive video display is disposed in front of the mechanical reel display to portray a video image superimposed upon electro-mechanical reels. In another combination disclosed in U.S. Pat. No. 7,654,899, a projector projects video images onto stationary or moving surfaces. In yet another combination disclosed in U.S. Pat. No. 7,452,276, miniature video displays are mounted to electro-mechanical reels and portray video symbols for the game. In a further combination disclosed in U.S. Pat. No. 8,591,330, flexible displays such as OLED or e-paper displays are affixed to electro-mechanical reels. The aforementioned U.S. Pat. Nos. 6,517,433, 7,654,899, 7,452,276, and 8,591,330 are incorporated herein by reference in their entireties.

The presentation devices **18**, **20**, the audio speakers **22**, lighting assemblies, and/or other devices associated with presentation are collectively referred to as a "presentation assembly" of the gaming machine **10**. The presentation assembly may include one presentation device (e.g., the primary presentation device **18**), some of the presentation devices of the gaming machine **10**, or all of the presentation devices of the gaming machine **10**. The presentation assembly may be configured to present a unified presentation sequence formed by visual, audio, tactile, and/or other suitable presentation means, or the devices of the presentation assembly may be configured to present respective presentation sequences or respective information.

The presentation assembly, and more particularly the primary presentation device **18** and/or the secondary presentation device **20**, variously presents information associated with wagering games, non-wagering games, community games, progressives, advertisements, services, premium entertainment, text messaging, emails, alerts, announcements, broadcast information, subscription information, etc. appropriate to the particular mode(s) of operation of the gaming machine **10**. The gaming machine **10** may include a touch screen(s) **24** mounted over the primary or secondary presentation devices, buttons **26** on a button panel, a bill/ticket acceptor **28**, a card reader/writer **30**, a ticket dispenser **32**, and player-accessible ports (e.g., audio output jack for headphones, video headset jack. USB port, wireless transmitter/receiver, etc.). It should be understood that numerous other peripheral devices and other elements exist and are readily utilizable in any number of combinations to create various forms of a gaming machine in accord with the present concepts.

The player input devices, such as the touch screen **24**, buttons **26**, a mouse, a joystick, a gesture-sensing device, a voice-recognition device, and a virtual-input device, accept player inputs and transform the player inputs to electronic data signals indicative of the player inputs, which correspond to an enabled feature for such inputs at a time of activation (e.g., pressing a "Max Bet" button or soft key to indicate a player's desire to place a maximum wager to play the wagering game). The inputs, once transformed into electronic data signals, are output to game-logic circuitry for processing. The electronic data signals are selected from a group consisting essentially of an electrical current, an electrical voltage, an electrical charge, an optical signal, an optical element, a magnetic signal, and a magnetic element.

The gaming machine **10** includes one or more value input/payment devices and value output/payout devices. In order to deposit cash or credits onto the gaming machine **10**, the value input devices are configured to detect a physical item associated with a monetary value that establishes a credit balance on a credit meter such as the "credits" meter **200** (see FIG. **4**). The physical item may, for example, be currency bills, coins, tickets, vouchers, coupons, cards, and/or computer-readable storage mediums. The deposited cash or credits are used to fund wagers placed on the wagering game played via the gaming machine **10**. Examples of value input devices include, but are not limited to, a coin acceptor, the bill/ticket acceptor **28**, the card reader/writer **30**, a wireless communication interface for reading cash or credit data from a nearby mobile device, and a network interface for withdrawing cash or credits from a remote account via an electronic funds transfer. In response to a cashout input that initiates a payout from the credit balance on the "credits" meter **200** (see FIG. **4**), the value output devices are used to dispense cash or credits from the gaming machine **10**. The credits may be exchanged for cash at, for example, a cashier or redemption station. Examples of value output devices include, but are not limited to, a coin hopper for dispensing coins or tokens, a bill dispenser, the card reader/writer **30**, the ticket dispenser **32** for printing tickets redeemable for cash or credits, a wireless communication interface for transmitting cash or credit data to a nearby mobile device, and a network interface for depositing cash or credits to a remote account via an electronic funds transfer.

Turning now to FIG. **2**, there is shown a block diagram of the gaming-machine architecture. The gaming machine **10** includes game-logic circuitry **40** securely housed within a locked box inside the gaming cabinet **12** (see FIG. **1**). The game-logic circuitry **40** includes a central processing unit (CPU) **42** connected to a main memory **44** that comprises one or more memory devices. The CPU **42** includes any suitable processor(s), such as those made by Intel and AMD. By way of example, the CPU **42** includes a plurality of microprocessors including a master processor, a slave processor, and a secondary or parallel processor. Game-logic circuitry **40**, as used herein, comprises any combination of hardware, software, or firmware disposed in or outside of the gaming machine **10** that is configured to communicate with or control the transfer of data between the gaming machine **10** and a bus, another computer, processor, device, service, or network. The game-logic circuitry **40**, and more specifically the CPU **42**, comprises one or more controllers or processors and such one or more controllers or processors need not be disposed proximal to one another and may be located in different devices or in different locations. The game-logic circuitry **40**, and more specifically the main memory **44**, comprises one or more memory devices which need not be disposed proximal to one another and may be located in different devices or in different locations. The game-logic circuitry **40** is operable to execute all of the various gaming methods and other processes disclosed herein. The main memory **44** includes a wagering-game unit **46**. In one embodiment, the wagering-game unit **46** causes wagering games to be presented, such as video poker, video blackjack, video slots, video lottery, etc., in whole or part.

The game-logic circuitry **40** is also connected to an input/output (I/O) bus **48**, which can include any suitable bus technologies, such as an AGTL+ frontside bus and a PCI backside bus. The I/O bus **48** is connected to various input devices **50**, output devices **52**, and input/output devices **54** such as those discussed above in connection with FIG. **1**. The I/O bus **48** is also connected to a storage unit **56** and an external-system interface **58**, which is connected to external system(s) **60** (e.g., wagering-game networks).

The external system **60** includes, in various aspects, a gaming network, other gaming machines or terminals, a gaming server, a remote controller, communications hardware, or a variety of other interfaced systems or components, in any combination. In yet other aspects, the external system **60** comprises a player's portable electronic device (e.g., cellular phone, electronic wallet, etc.) and the external-system interface **58** is configured to facilitate wireless communication and data transfer between the portable electronic device and the gaming machine **10**, such as by a near-field communication path operating via magnetic-field induction or a frequency-hopping spread spectrum RF signals (e.g., Bluetooth, etc.).

The gaming machine **10** optionally communicates with the external system **60** such that the gaming machine **10** operates as a thin, thick, or intermediate client. The game-logic circuitry **40**—whether located within ("thick client"), external to ("thin client"), or distributed both within and external to ("intermediate client") the gaming machine **10**—is utilized to provide a wagering game on the gaming machine **10**. In general, the main memory **44** stores programming for a random number generator (RNG), game-outcome logic, and game assets (e.g., art, sound, etc.)—all of which obtained regulatory approval from a gaming control board or commission and are verified by a trusted authentication program in the main memory **44** prior to game execution. The authentication program generates a live authentication code (e.g., digital signature or hash) from the memory contents and compare it to a trusted code stored in the main memory **44**. If the codes match, authentication is deemed a success and the game is permitted to execute. If, however, the codes do not match, authentication is deemed a failure that must be corrected prior to game execution. Without this predictable and repeatable authentication, the gaming machine **10**, external system **60**, or both are not allowed to perform or execute the RNG programming or game-outcome logic in a regulatory-approved manner and are therefore unacceptable for commercial use. In other words, through the use of the authentication program, the game-logic circuitry facilitates operation of the game in a way that a person making calculations or computations could not.

When a wagering-game instance is executed, the CPU **42** (comprising one or more processors or controllers) executes the RNG programming to generate one or more pseudo-random numbers. The pseudo-random numbers are divided into different ranges, and each range is associated with a respective game outcome. Accordingly, the pseudo-random numbers are utilized by the CPU **42** when executing the game-outcome logic to determine a resultant outcome for that instance of the wagering game. The resultant outcome is then presented to a player of the gaming machine **10** by accessing the associated game assets, required for the resultant outcome, from the main memory **44**. The CPU **42** causes the game assets to be presented to the player as outputs from the gaming machine **10** (e.g., audio and video presentations). Instead of a pseudo-RNG, the game outcome may be derived from random numbers generated by a physical RNG that measures some physical phenomenon that is expected to be random and then compensates for possible biases in the measurement process. Whether the

RNG is a pseudo-RNG or physical RNG, the RNG uses a seeding process that relies upon an unpredictable factor (e.g., human interaction of turning a key) and cycles continuously in the background between games and during game play at a speed that cannot be timed by the player. Accordingly, the RNG cannot be carried out manually by a human and is integral to operating the game.

The gaming machine **10** may be used to play central determination games, such as electronic pull-tab and bingo games. In an electronic pull-tab game, the RNG is used to randomize the distribution of outcomes in a pool and/or to select which outcome is drawn from the pool of outcomes when the player requests to play the game. In an electronic bingo game, the RNG is used to randomly draw numbers that players match against numbers printed on their electronic bingo card.

The gaming machine **10** may include additional peripheral devices or more than one of each component shown in FIG. **2**. Any component of the gaming-machine architecture includes hardware, firmware, or tangible machine-readable storage media including instructions for performing the operations described herein. Machine-readable storage media includes any mechanism that stores information and provides the information in a form readable by a machine (e.g., gaming terminal, computer, etc.). For example, machine-readable storage media includes read only memory (ROM), random access memory (RAM), magnetic-disk storage media, optical storage media, flash memory, etc.

In accordance with various methods of conducting a wagering game on a gaming system in accord with the present concepts, the wagering game includes a game sequence in which a player makes a wager, and a wagering-game outcome is provided or displayed in response to the wager being received or detected. The wagering-game outcome, for that particular wagering-game instance, is then revealed to the player in due course following initiation of the wagering game. The method comprises the acts of conducting the wagering game using a gaming apparatus, such as the gaming machine **10** depicted in FIG. **1**, following receipt of an input from the player to initiate a wagering-game instance. The gaming machine **10** then communicates the wagering-game outcome to the player via one or more output devices (e.g., primary presentation device **18** or secondary presentation device **20**) through the presentation of information such as, but not limited to, text, graphics, static images, moving images, etc., or any combination thereof. In accord with the method of conducting the wagering game, the game-logic circuitry **40** transforms a physical player input, such as a player's pressing of a "Spin" touch key or button, into an electronic data signal indicative of an instruction relating to the wagering game (e.g., an electronic data signal bearing data on a wager amount).

In the aforementioned method, for each data signal, the game-logic circuitry **40** is configured to process the electronic data signal, to interpret the data signal (e.g., data signals corresponding to a wager input), and to cause further actions associated with the interpretation of the signal in accord with stored instructions relating to such further actions executed by the controller. As one example, the CPU **42** causes the recording of a digital representation of the wager in one or more storage media (e.g., storage unit **56**), the CPU **42**, in accord with associated stored instructions, causes the changing of a state of the storage media from a first state to a second state. This change in state is, for example, effected by changing a magnetization pattern on a magnetically coated surface of a magnetic storage media or changing a magnetic state of a ferromagnetic surface of a magneto-optical disc storage media, a change in state of transistors or capacitors in a volatile or a non-volatile semiconductor memory (e.g., DRAM, etc.). The noted second state of the data storage media comprises storage in the storage media of data representing the electronic data signal from the CPU **42** (e.g., the wager in the present example). As another example, the CPU **42** further, in accord with the execution of the stored instructions relating to the wagering game, causes the primary presentation device **18**, other presentation device, or other output device (e.g., speakers, lights, communication device, etc.) to change from a first state to at least a second state, wherein the second state of the primary presentation device comprises a visual representation of the physical player input (e.g., an acknowledgement to a player), information relating to the physical player input (e.g., an indication of the wager amount), a game sequence, an outcome of the game sequence, or any combination thereof, wherein the game sequence in accord with the present concepts comprises acts described herein. The aforementioned executing of the stored instructions relating to the wagering game is further conducted in accord with a random outcome (e.g., determined by the RNG) that is used by the game-logic circuitry **40** to determine the outcome of the wagering-game instance. In at least some aspects, the game-logic circuitry **40** is configured to determine an outcome of the wagering-game instance at least partially in response to the random parameter.

In one embodiment, the gaming machine **10** and, additionally or alternatively, the external system **60** (e.g., a gaming server), means gaming equipment that meets the hardware and software requirements for fairness, security, and predictability as established by at least one state's gaming control board or commission. Prior to commercial deployment, the gaming machine **10**, the external system **60**, or both and the casino wagering game played thereon may need to satisfy minimum technical standards and require regulatory approval from a gaming control board or commission (e.g., the Nevada Gaming Commission, Alderney Gambling Control Commission, National Indian Gaming Commission, etc.) charged with regulating casino and other types of gaming in a defined geographical area, such as a state. By way of non-limiting example, a gaming machine in Nevada means a device as set forth in NRS 463.0155, 463.0191, and all other relevant provisions of the Nevada Gaming Control Act, and the gaming machine cannot be deployed for play in Nevada unless it meets the minimum standards set forth in, for example, Technical Standards 1 and 2 and Regulations 5 and 14 issued pursuant to the Nevada Gaming Control Act. Additionally, the gaming machine and the casino wagering game must be approved by the commission pursuant to various provisions in Regulation 14. Comparable statutes, regulations, and technical standards exist in or are used in other gaming jurisdictions, including for example GLI Standard #11 of Gaming Laboratories International (which defines a gaming device in Section 1.5) and N.J.S.A 5:12-23, 5:12-45, and all other relevant provisions of the New Jersey Casino Control Act. As can be seen from the description herein, the gaming machine **10** may be regulatorily approved and thus implemented with hardware and software architectures, circuitry, and other special features that differentiate it from general-purpose computers (e.g., desktop PCs, laptops, and tablets).

Referring now to FIGS. **3A-3B**, there is shown a flow diagram representing one data processing method corresponding to at least some instructions stored and executed by the game-logic circuitry **40** in FIG. **2** to perform operations according to an embodiment of the present invention.

The data processing method is described below in connection with the exemplary presentations of different spin outcomes in FIGS. **4-10**.

Game Play Initiation

Referring to FIG. **3**A, the data processing method commences at step **300**. At step **302**, the game-logic circuitry controls one or more presentation devices (e.g., mechanical-reel display device, video display device, or a combination thereof) to present a plurality of symbol-bearing reels, an array of symbol positions, and a plurality of persistent elements. Although the method is described with respect to one presentation device, it is to be understood that the presentation described herein may be performed by a presentation assembly including more than one presentation device. The symbol positions of the array may be arranged in a variety of configurations, formats, or structures and may comprise a plurality of rows and columns. The rows of the array are oriented in a generally horizontal direction, and the columns of the array are oriented in a generally vertical direction. The symbol positions in each row of the array are horizontally aligned with each other, and the symbol positions in each column of the array are vertically aligned with each other. Alternatively, the symbol positions may be arranged in a honeycomb configuration with adjacent columns vertically offset from each other by one-half symbol position or adjacent rows horizontally offset from each other by one-half symbol position. The number of symbol positions in different rows and/or different columns may vary from each other. The reels may be associated with the respective columns of the array such that the reels spin vertically, and each reel populates a respective column. In another embodiment, the reels may be associated with the respective rows of the array such that the reels spin horizontally, and each reel populates a respective row. In some embodiments, the reels are associated with respective individual symbol positions of the array such that each reel animates in place and populates only its respective symbol position. The symbol array configuration may vary between the base game and any bonus games utilizing the array.

Referring to FIGS. **4-10**, which illustrate examples of the display at the conclusion of various representative game spins, the symbol array **210** in the base game has a three-by-five rectangular configuration, and each symbol position is associated with a respective independent reel. The reels bear a plurality of symbols that may, for example, include various base game symbols 10, J, Q, K. A and a WILD symbol that can substitute for any of the base game symbols. The plurality of symbols may also include value-bearing symbols. An example of landed base game symbols and a two value-bearing symbols bearing the values 150 and 200 are shown in FIG. **4**.

FIG. **5** illustrates that the plurality of symbols also includes accumulation symbols in the form of BLUE, GOLD and RED coin symbols. Players hope to land the BLUE, GOLD or RED coin symbols in the symbol array **210** in order to enter a bonus mode supplemented by special rules associated with three persistent elements **212**, **214**, **216**. In the example of FIG. **5**, a BLUE COIN is associated with the first persistent element **212**, a GOLD coin is associated with the second persistent element **214**, and a RED coin is associated with the third persistent element **216**. In the examples of FIGS. **4-10**, a "Frenzy" game feature is associated with the first persistent element **212**, a "3× Wild" game feature is associated with the second persistent element **214**, and an "Instant Win" game feature is associated with the third persistent element **216**. These game features are examples. In other embodiments, different game features of various types may be associated with the persistent elements **212**, **214** and **216**, as will be described further below. In some embodiments, the accumulation symbols may also be value-bearing symbols. For example, a value bearing symbol, such as those illustrated in FIG. **4**, may also be colored BLUE, GOLD or RED to associate it with one of the persistent elements **212**, **214** and **216**.

In FIGS. **4-10**, the persistent elements are represented as triangles. The persistent elements **212**, **214** and **216** may take forms other than triangles, including, for example, coin pots, urns, vases, jars, jugs, cans, bowls, piggy banks, beehives, inflating balloons, ladders, dials, meters, etc. Similarly, the accumulation symbols associated with the persistent elements may take forms other than coins, including, for example, balloons, colored dollar signs, etc. The accumulation symbols may or may not be color-matched to their respective persistent elements, provided their association to a persistent element is indicated. For example, the BLUE coins and the first persistent element **212** may be labeled or colored blue, the GOLD coins and the second persistent element **214** may be labeled or colored gold, and the RED coins and the third persistent element **216** may be labeled or colored red.

At step **304**, the game-logic circuitry detects, via a value input device, a physical item associated with a monetary value that establishes a monetary balance in the form of cash or credits. In FIGS. **4-10**, the monetary balance may be shown on a meter **200**.

Game Spin

At step **306**, the game-logic circuitry initiates a game of a wagering game cycle (i.e., spin cycle) in response to an input indicative of a wager covered by the monetary balance. To initiate a spin of the reels, the player may press a "Spin" or "Max Bet" key on a button panel or touch screen. In FIGS. **4-10** the wager may be shown on a bet meter **202**.

At step **308**, using an RNG, the game-logic circuitry spins and stops the reels to randomly land symbols from the reels in the array in visual association with one or more paylines (also known as lines, ways, patterns, or arrangements). The reel spin may be animated on a video display by depicting symbol-bearing strips moving vertically across the display and synchronously updating the symbols visible on each strip as the strip moves across the display. Alternatively, the reels may be physical/electromechanical reels. FIGS. **4-10** depict various game spin outcomes.

At step **310**, the game-logic circuitry determines whether or not any accumulation (coin) symbols landed in the array **210** and, if not, proceeds to step **312**, to be described later. If the game-logic circuitry determines that at least one accumulation symbol landed in the array, flow proceeds to step **324** of FIG. **3**B. This example will be described first.

Animation of Persistent Element Growth

FIG. **6** illustrates the results of a game spin in which accumulation coins have landed in the array. At step **324**, the game-logic circuitry animates, via the one or more presentation devices, an addition of any coin in the array to the persistent element associated with the coin. To show the transfer of a coin symbol to a persistent element, the coin may be animated to "fly" from the array to land in its associated persistent element. To represent the gradual addition of coins to the persistent element, the persistent element and/or the volume of coins therein may appear to grow in size. In accordance with some embodiments, the persistent element may be an object of fixed size accompanied by some other indication of accumulating value, for example, by a gradual change in color shading, for example, from light red to dark red. In other embodiments, the persistent element may change size and also show some other indication of increasing value, for example, the color of the display in the immediate area of the persistent element may gradually change as its value increases. The size of a persistent element may or may not indicate the likelihood that its associated game feature will be triggered. This will be discussed further below.

In one or more embodiments, the coin symbols may be carried on the reels as "stickers" applied over an underlying standard reel symbol. As part of the animation sequence at step **324**, in which the coin is transferred to its respective persistent element, the coin "sticker" is removed from its location in the reel array to reveal the standard reel symbol underneath. In other embodiments, the coin symbols may simply be removed from the array and replacement symbols may be randomly selected to take their places. If the coin symbols are also value-bearing symbols, as mentioned above, the value-bearing coin symbols may remain in the array and contribute their values to the game's outcome. In other embodiments, non-value-bearing coin symbols may remain in the array as "blocking symbols" that break up other potentially winning combinations of standard symbols. These approaches or combinations thereof all fall within the spirit and scope of the invention.

In FIG. **6**, each transfer of a coin to the persistent elements **212**, **214**, **216** is represented by a "coin" circle and an arrow. See, for example, the arrow from the BLUE coin in the second column of the array **210**, the arrow from the GOLD coin in the fourth column of the array **210**, and the arrow from the RED coin in the fifth column of the array **210** in FIG. **7**. Though not shown in FIG. **7**, no coins, coins of only one or two colors or multiple coins of the same color/type may appear in the array at the same time. Animation representing the transfer each coin from the array **210** to its respective persistent element may be presented sequentially or in parallel.

Game Feature Triggering

At step **326**, the game-logic circuitry randomly determines, via the RNG, whether or not to trigger the game features associated with any coin symbols detected at step **310**. This random determination is independent of any prior wagering game cycles. In accordance with one or more embodiments, if any coin symbols appear in the array, the odds of triggering the game features at step **326** may increase according to the number of coins appearing in the array. This may be accomplished, for example, by changing weights associated with the random determination. In other embodiments, the appearance of multiple coins in the array has no effect on the probability of awarding the game features associated with the persistent elements corresponding to the coins appearing in the array. If the game features are triggered, for each coin symbol detected at step **310**, an indication such as a "WINNER!" label may appear next to each respective coin's associated persistent element, or the respective persistent element may be otherwise highlighted. Persistent elements associated with non-triggered game features may be further deemphasized, for example, by shading or "greying" them out. FIG. **7** illustrates an example of a game spin in which only the GOLD game feature has been won.

If a game feature is not triggered at step **326**, the game-logic circuitry returns to FIG. **3**A and proceeds to step **312**, described further below. If, however, at least one game feature was triggered at step **326**, the game-logic circuitry instead proceeds to step **328** to begin the process of awarding one or more game features.

In some embodiments, a "game feature" simply awards a prize, for example a fixed or progressive jackpot amount. In some embodiments, a game feature may comprise any type of bonus game. Non-limiting examples of bonus games include a certain number of free games (i.e., spins of the reels), a "pick'em" bonus game, a wheel-spinning game, etc. In still other embodiments, a bonus game may be played to determine payment of prize, for example, a fixed or progressive jackpot. In some embodiments, a game feature may comprise an enhancement to the game. An enhancement may include, without limitation, pay table modifiers such as multipliers, increased values on value-bearing symbols, modification to the reels to include improved symbols, such as wild symbols, modification of symbol weights or the removal of certain "blocking symbols", additional rows or columns added to the array, additional free spins, replacement symbols for symbols already present in the array, etc. The enhancement may be applied to the current base game outcome or may be applied to one or more subsequent bonus games or wagered base game plays, as will be shown. In some embodiments, a game feature may provide a second plurality of symbol-bearing reels and a second array of symbol positions to be filled, as described above, for a certain number of game cycles. Each game feature has a different impact on the expected value (EV) of the game. The relative frequency of the game features may be controlled by adjusting which coin pot they are associated with in the cascade sequence, for example. Thus, a less lucrative game feature may be won more frequently, for example, approximately once in every ten game spin cycles, while a higher paying game feature may only occur approximately once in every one hundred game spin cycles according to a weighted cascade value determination. This allows game features to be triggered more often, for player enjoyment, while maintaining the overall expected EV of the wagering game.

Perceived Persistence

During a player's gaming session, the growth in size of the persistent elements **212**, **214** and **216** persists from one wagering game cycle to the next such that the player perceives that a game feature corresponding to a persistent element may be getting closer to being triggered. When the size of the persistent elements has no bearing on whether the associated game feature will actually be triggered, this is known as "perceived persistence." When a game feature associated with a persistent element is triggered, at least some of the contents of the persistent element are visually removed and the accumulation of value in that persistent element during subsequent game spin cycles resumes from that point. Each of the persistent elements **212**, **214** and **216** exhibits perceived persistence.

At step **328**, if at least one BLUE coin associated with the first persistent element **212** was present in the array at step **310**, the game logic circuitry awards the game feature associated with the first persistent element **212** at step **330**, otherwise the game-logic circuitry continues the flow at step **332**.

In the examples shown in FIGS. **4**-**10**, the game feature associated with the first persistent element **212** is "Frenzy." This exemplar game feature increases the value of value-bearing symbols on the reels for the duration of a bonus mode to be initiated at step **340**, below, once any other game features have been awarded. The reel strip layouts may otherwise remain the same as those of the base game or may be different. Once the "Frenzy" game feature has been awarded at step **330**, the game logic circuitry proceeds to step **332**.

At step **332**, if at least one GOLD coin associated with the second persistent element **214** was present in the array at step **310**, the game logic circuitry awards the game feature associated with the second persistent element **214** at step **334**, otherwise the game-logic circuitry continues the flow at step **336**.

In the example shown, the game feature associated with the second persistent element **214** is "3× Wilds," in which the wilds symbols on the reels will be replaced by 3× Wild symbols for the duration of a bonus mode to be initiated at step **340**, below, once any other game features have been awarded. The reel strip layouts may otherwise remain the same as those of the base game or may be different. Once the "3× Wild" game feature has been awarded at step **334**, the game logic circuitry proceeds to step **336**.

At step **336**, if at least one RED coin associated with the third persistent element **216** was present in the array at step **310**, the game logic circuitry awards the game feature associated with the third persistent element **216** at step **338**, otherwise the game-logic circuitry continues the flow at step **340**.

In the example shown, the game feature associated with the third persistent element **216** is an "Instant Win" game feature, in which a special "catalyst" symbol is added to one or more of the reels for the duration of a bonus mode to be initiated at step **340**, below. When the catalyst symbol lands in the array **210**, the values on any value-bearing symbols in the array **210** are awarded. This is advantageous to the player because, unless a certain number of value-bearing symbols appear in the array **210**, for example, six, the values on the symbols are not awarded. When less than the required number of value-bearing symbols are present, they tend to serve a "blocking symbols," breaking up other potential winning combinations. The reel strip layouts may otherwise remain the same as those of the base game or may be different. Once the game feature has been awarded at step **338**, the game logic circuitry proceeds to step **340**.

Applying Game Features/Bonus Mode Initiation

At step **340**, once all of the eligible game features have been awarded at one or more of steps **330**, **334** and **338**, if not already in a bonus mode, the game is placed in a bonus mode for a certain number of spins, determined via the RNG. In some embodiments, the number of spins is predetermined, for example, five spins. A bonus mode spin counter **220** associated with the bonus mode may be initialized and displayed, (FIG. **9**) The bonus mode spin counter will later be modified and tested at step **316**, described below. The bonus mode games will be played according to game rules and conditions set by the one or more awarded game features, enhancing the excitement and EV of the bonus mode games. For example, if only the "Frenzy" game feature was awarded at step **330**, the reels will have enhanced value-bearing symbols for the duration of the bonus games. Likewise, if only the "3× Wild" was awarded at step **334**, the reels will be modified with 3× Wild symbols for the duration of the bonus mode games. Finally, if only the "Instant Win" game feature was awarded at step **338**, the reels will be enhanced with one or more catalyst symbols for the duration of the bonus mode. It should be noted that multiple game features may be awarded if the spin that triggered the awarding of game features and the initiation of the bonus mode resulted in coins of more than one color appearing simultaneously in the array **210**. In the case where all three game features have been awarded, the reels will be enhanced by increasing the values of the value-bearing symbols, Wild symbols will become 3× Wild symbols and one or more catalyst symbols will also be added to one or more reels.

In the exemplar embodiment, once the bonus mode is initialized at step **340**, the coin symbols will be removed from the reel strips for the duration of the bonus mode. At the conclusion of the bonus mode, any pay rule or game modifications will be reset to the standard base game configuration at step **318**, described below. The game logic circuitry then returns to FIG. **3A**, proceeding to step **312**.

Win Evaluation

Once any animation of persistent element growth has occurred, any game features have been awarded and any bonus mode initiated, the game-logic circuitry evaluates the symbols in the array at step **312**. Payouts are awarded in accordance with either base game pay rules or bonus mode game rules, depending on whether the bonus mode is in effect for the current spin. The pay rules may include a pay table. The pay table may, for example, include "line pays," "ways pays" and "scatter pays." Line pays occur when a predetermined type and number of symbols appear along an activated payline, typically in a particular order such as left to right, right to left, top to bottom, bottom to top, etc. Ways pays appear on adjacent reels without the requirement to be on a specified pay line or directly adjacent to one another. Scatter pays occur when a predetermined type and number of symbols appear anywhere in the displayed array without regard to position or paylines. Each payline preferably consists of a single symbol position in each column of the array. The number of paylines may be as few as one or as many as possible given each payline consists of a single symbol position in each column of the array. To animate a standard pay, the display may apply a border, pattern, color change, background change, watermark, or other distinguishing characteristic to the winning payline and/or winning symbols that contributed to the pay. FIG. **8**, for example, depicts a line pay of three K symbols in the bottom row of the array **210**. The awarded pay is added to a win meter **204**.

The pay rules may include the awarding of one or more free spins. In one example, a certain number of free spins, for example, five free spins, may be awarded by the game rules if six or more value-bearing symbols land in the array **210**. FIG. **10**, described further below, illustrates an example of a game outcome that triggers free spins. In some embodiments, the number of free spins is predetermined. In other embodiments, the number of free spins may be randomly determined and vary between each awarding of free spins. The free spins may be played as a "hold and spin" game in which certain types of symbols, once landed in the array, are held in place and may variously persist in the array for at least one additional free spin cycle, until they contribute to a winning outcome, until all of the free spins have been played, etc. These persistent symbols typically include value-bearing symbols, which may or may not be awarded until a certain number of the value-bearing symbols have been held in the array. In some embodiments, wild symbols or other symbols that may improve the chances of winning or provide higher pays may also be held.

At step **314**, the game-logic circuitry checks to see whether the game is in a free spin state as a result of free games having been previously awarded at step **312**. If the game is in a free game state, the game logic circuitry returns to step **308** for another spin without requiring a wager.

Bonus Mode Termination

If the game in not in a free game state, that is, if the free spin counter **230** is, for example, zero, the game-logic circuitry advances to step **316** to check whether any bonus mode in effect should be canceled. As part of this evaluation, step **316** may modify the bonus mode spin counter **220** (FIG. **9**), for example, by subtracting one from the number of remaining bonus mode spins prior to testing for bonus mode termination. If bonus mode termination is required, at step **318**, the game-logic circuitry removes all game rules previously applied at step **340** and the bonus mode counter **220** is removed.

Game Play Continuation/Termination

At step **320**, the game-logic circuitry determines whether or not it has received a cashout input via at least one of the one or more player input devices of the gaming machine. If it has not received a cashout input, the game-logic circuitry waits for the next wager input at step **306**. If it has received a cashout input, the game-logic circuitry initiates a payout from the monetary balance on the meter **200** in FIGS. **4-10**. The data processing method then ends at step **322**.

Examples of Spins Related to Method **300**

In accordance with one or more embodiments, FIG. **9** illustrates an example of the conclusion of a game spin while in the bonus mode with the "Frenzy," "3× Wilds" and "Instant Win" game features in effect. Because the "Frenzy" game feature is in effect, the value-bearing symbols bear larger values, 1200 and 2000, compared to the value bearing symbols shown in FIGS. **4** and **6**. Because the "3× Wilds" game feature is in effect, the wild symbols bear 3× multipliers. Finally, a catalyst symbol appears in the array as a result of the "Instant Win" game feature being in effect. While the 3× Wild symbols do not create any winning combinations with the other symbols in the array, the game will award 3200 credits (1200+2000) for the "Instant Win" game feature when wins are evaluated at step **312**.

In accordance with one or more embodiments, FIG. **10** illustrates an example of the conclusion of a game spin that triggers free games. In this example, six value-bearing symbols appear in the array, which, according to the exemplar game rules described above, awards five free spins at step **312**. The number of free spins remaining is reflected in a free spin meter **230**. Note that none, one, two or three of the game features may also be in effect during the free games, depending on whether the game is already in the bonus mode or not. In the example of FIG. **10**, the game is not in a bonus mode, thus the bonus mode spin counter **220** (FIG. **9**) is not displayed.

Referring now to FIGS. **11A-11**B, there is shown a flow diagram representing a second data processing method corresponding to at least some instructions stored and executed by the game-logic circuitry **40** in FIG. **2** to perform operations according to an embodiment of the present invention. The data processing method is also described in connection with the exemplary presentations of different spin outcomes in FIGS. **4-10**.

Game Play Initiation

Referring to FIG. **11**A, the data processing method commences at step **1100**. At step **1102**, the game-logic circuitry controls one or more presentation devices (e.g., mechanical-reel display device, video display device, or a combination thereof) to present a plurality of symbol-bearing reels, an array of symbol positions, and a plurality of persistent elements as described above with respect to step **302**.

At step **1104**, the game-logic circuitry detects, via a value input device, a physical item associated with a monetary value that establishes a monetary balance in the form of cash or credits as described above with respect to step **304**.

Game Spin

At step **1106**, the game-logic circuitry initiates a game of a wagering game cycle (i.e., spin cycle) in response to an input indicative of a wager covered by the monetary balance as described above with respect to step **306**.

At step **1108**, using an RNG, the game-logic circuitry spins and stops the reels to randomly land symbols from the reels in the array as described above with respect to step **308**.

At step **1110**, the game-logic circuitry determines whether or not any accumulation (coin) symbols landed in the array **210** and, if not, proceeds to step **1112**, to be described later. If the game-logic circuitry determines that at least one accumulation symbol landed in the array, flow proceeds to step **1124** of FIG. **11**B. This example will be described first.

Animation of Persistent Element Growth

At step **1124**, the game-logic circuitry animates, via the one or more presentation devices, an addition of any coin in the array to the persistent element associated with the coin as described above with respect to step **324**.

Game Feature Triggering

At step **1126**, the game-logic circuitry randomly determines, via the RNG, whether or not to award the game feature associated with any BLUE coin symbols in array **210**. Unlike the previously described embodiment in which a single decision is made to award all game features having an associated coin in array **210** (at step **326**), this embodiment independently determines whether to award only the game feature associated with the BLUE coin at step **1126**.

If the decision is made to not award the BLUE game feature, flow proceeds to step **1130**, otherwise, the associated game feature is awarded at step **1128** in the manner described with respect to step **330**, above. In addition, if the game is not currently in a bonus mode, the game is placed in a bonus mode for a certain number of spins, randomly determined via the RNG. In some embodiments, the number of spins is predetermined; for example, five spins. As described above, a spin counter **220** associated with the bonus mode may be initialized and displayed (FIG. **9**). This spin counter will be modified and tested at step **1116**, described below.

If the game is already in the bonus mode, the bonus mode is extended by the number of spins determined at step **1128** and the spin counter **220** is increased by the new number of spins. Thus, awarding of the BLUE persistent element either initiates a new bonus mode or extends an existing bonus mode with the addition of the BLUE game feature, in this case, the "Frenzy" rules, applied to the game rules currently in effect.

At step **1130**, the game-logic circuitry randomly determines, via the RNG, whether or not to award the game feature associated with any GOLD coin symbols in array **210**. Again, the game-logic circuitry independently determines whether or not to award the game feature associated with the GOLD persistent element at this step.

If the decision is made to not award the GOLD game feature, flow proceeds to step **1134**, otherwise, the associated game feature is awarded at step **1132** in the manner described with respect to step **334**, above. As above, if the game is not currently in a bonus mode, the game is placed in a bonus mode for a certain number of spins, as described above with respect to step **1128**. As described above, the bonus mode counter **220** may be initialized and displayed (FIG. **9**). This spin counter will be modified and tested at step **1116**, described below.

If the game is already in the bonus mode, the bonus mode is extended by the number of spins determined at step **1132** and the bonus mode counter **220** is increased by the new number of spins. Whether the awarding of the GOLD persistent element initiates a new bonus mode or extends an existing bonus mode, the "3× Wilds" GOLD game feature rules will be applied to the game rules currently in effect.

At step **1134**, the game-logic circuitry randomly determines, via the RNG, whether or not to award the game feature associated with any RED coin symbols in array **210**. At this step, the game independently determines whether to award only the game feature associated with the RED persistent element. If the decision is made to not award the RED game feature, flow proceeds to step **1138**, otherwise, the associated game feature is awarded at step **1136** in the manner described with respect to step **338**, above.

Once again, if the game is not currently in a bonus mode, the game is placed in a bonus mode for a certain number of spins, as described above with respect to step **1128**. Also as described above, the bonus mode counter **220** may be initialized and displayed (FIG. **9**). This spin counter will be modified and tested at step **1116**, described below.

If the game is already in a bonus mode, the bonus mode is extended by the number of spins determined at step **1136** and the bonus mode counter **220** is increased by the new number of spins. Whether the awarding of the RED persistent element initiates a new bonus mode or extends an existing bonus mode with the addition of the RED game feature, the "Instant Win" rules will be applied to the game rules currently in effect.

Applying Game Features/Bonus Mode Initiation/Extension

At step **1138**, any newly awarded game features from steps **1128**, **1132** or **1136** are applied to the current game rules in effect as described with respect to step **330**, above. Thus, the bonus mode games will be played according to game rules and conditions set by previously awarded game features enhanced by any newly awarded game features. Multiple game features may be in effect at the same time, as described above with reference to step **340**.

If multiple game features were newly awarded in steps **1126**, **1130** or **1134**, each additional extension will be applied to the bonus mode counter **220**. The spin counter **220** will reflect the cumulative effect of the previous value of the bonus mode counter **220** plus any newly awarded spins. For example, if the bonus mode counter **220** value was 0 at the step **1138** (signifying that no bonus mode was in effect) and 4 spins were awarded at step **1126**, 6 spins awarded at step **1130**, and 4 spins awarded at step **1134**, the bonus mode counter **220** would be initialized to reflect 14 (4+6+4) bonus mode spins and the bonus mode would be initialized accordingly.

In another example, if a bonus mode was already in effect from a previous game cycle and the bonus mode counter value was 7, 4 additional spins awarded at step **1126** would set the bonus mode counter **220** to 11 (7+4).

Once any newly awarded game features have been applied to the game rules and any updates to the bonus game spin counter **220** have been applied at step **1138**, the game-logic circuitry returns to FIG. **11**A and proceeds to step **1112**.

Win Evaluation

Once any animation of persistent element growth has occurred, any game features have been awarded and any bonus mode initiated or extended, the game-logic circuitry evaluates the symbols in the array at step **1112** as described above with reference to step **312**.

At step **1114**, the game-logic circuitry checks to see whether the game is in a free spin state as a result of free spins having been previously awarded at step **1112**. If the game is in a free spin state, the game logic circuitry returns to step **1108** for another spin without requiring a wager.

Bonus Mode Termination

If the game in not in a free spin state, the game-logic circuitry advances to step **1116** to check whether any bonus mode should be canceled, as described above with reference to step **316**. If a bonus mode should be canceled, at step **1118**, all game rules previously applied at step **1138** are removed and the bonus mode counter **220** is removed.

Game Play Continuation/Termination

At step **1120**, the game-logic circuitry determines whether or not it has received a cashout input via at least one of the one or more player input devices of the gaming machine. If it has not received a cashout input, the game-logic circuitry waits for the next wager input at step **1106**. If it has received a cashout input, the game-logic circuitry initiates a payout from the monetary balance on the meter **200** in FIGS. **4-10**. The data processing method then ends at step **1122**.

Examples of Spins Related to Method **1100**

Spin 1: In one example of a series of spins illustrating the concepts covered by method **1100**, a credit balance is established, and a wager is accepted to initiate a first spin. The game is in a base game state with no free games and is not in bonus mode.

The first spin results in no winning paytable outcomes, no awarding of free games and no awarding of game features. Another wager is accepted to initiate the next spin.

Spin 2: The second spin results in a winning paytable outcome only, which is awarded. FIG. **8** provides an example of such an outcome. Another wager is accepted to initiate the next spin.

Spin 3: The third spin, like the first, provides no winning outcomes. Another wager is accepted to initiate the next spin.

Spin 4: The fourth spin lands six value-bearing symbols in the array. The values on the value-bearing symbols are awarded and, according to the rules of the game, five free spins are also awarded and the free spin counter **230** is set to 5. FIG. **10** provides an example of such an outcome. No wager is required for the next spin because free spins have been awarded.

Spin 5: The fifth spin, like the first and third spins, produces no winning outcomes. The free spin counter **230** is decremented to 4. Free games remain, thus, no wager is required for the next spin.

Spin 6: A GOLD coin lands in the array on the sixth spin. The GOLD game feature is awarded and all wild symbols on the reels become 3× Wild symbols. Bonus mode is initiated and will be in effect for five spins. The bonus mode counter **220** is displayed and set to 5. FIG. **9** provides an example of such an outcome. The free spin counter **230** is decremented to 3. Because free games remain, no wager is required for the next spin.

Spin 7: The seventh spin results in a line pay win that includes a 3× Wild and the normal pay value is multiplied by three and awarded. The bonus mode counter **220** is decremented to 4, the free spin counter **230** is decremented to 2 and no wager is required for the next spin.

Spin 8: The eighth spin results in no winning outcomes. The bonus mode counter **220** is decremented to 3, the free spin counter **230** is decremented to 1 and no wager is required for the next spin.

Spin 9: The ninth spin results in no winning outcomes, though a BLUE, a GOLD and a RED coin all appear in the array. FIG. **6** provides an example of an outcome where coins land in the array and affect the size of their associated persistent elements without triggering the awarding of any game features. The bonus mode counter is decremented to 2 and the free spin counter is decremented to 0 and removed from the display. Because no free spins remain, a wager is required for the next spin. Even though the free spin period is complete, the 3× Wild symbols remain in effect for an additional two spins, paid or otherwise.

Spin 10: A BLUE coin lands in the array on the tenth spin. Awarding of the BLUE game feature ("Frenzy") is triggered and the values borne by all value-bearing symbols on the reels are increased. Bonus mode is extended by two spins and the bonus mode counter **220** is increased to 4. At this time, both the GOLD and the BLUE game features are applied to the game rules. Because no free spins are in effect, a wager is required for the next spin and is accepted.

Spin 11: The eleventh spin results in no winning outcomes. The bonus mode counter **220** is decremented to 3. A wager is required for the next spin and is accepted.

Spin 12: A RED coin lands in the array on the twelfth spin. Awarding of the RED game feature ("Instant Win") is triggered and the catalyst symbol is added to one or more reels, as described above. Bonus mode is extended by two spins and the bonus mode counter **220** is increased to 4. At this time, both the GOLD and the BLUE game features are applied to the game rules. Because no free spins are in effect, a wager is required for the next spin and is accepted.

Spin 13: On the thirteenth spin, two value-bearing symbols appear in the array along with the catalyst symbol. Because the "Instant Win" game feature is in effect, the values borne by the value-bearing symbols are awarded. For example, if the value-bearing symbols bore **1200** and **2000**, as shown in FIG. **9**, 3200 credits would be awarded. The bonus mode counter is decremented to 3. A wager is required for the next spin and is accepted.

Spin 14: The fourteenth spin results in no winning outcomes. The bonus mode counter **220** is decremented to 2. A wager is required for the next spin and is accepted.

Spin 15: The fifteenth spin results in two line pays, which are awarded. One of the line pays is multiplied by three because it contains a 3× Wild symbol. The bonus mode counter is decremented to 1. A wager is required for the next spin and is accepted.

Spin 16: The sixteenth spin results in no winning outcomes. The bonus mode counter **220** is decremented to 1. A wager is required for the next spin and is accepted.

Spin 17: The seventeenth spin results in no winning outcomes. The bonus mode counter is decremented to 0 and removed from the display. Bonus mode is terminated, and the effects of the "Frenzy" and "Instant Win" game features are removed from the game rules. The game returns to normal base game play. A wager is required for the next spin and is accepted.

Spin 18: The eighteenth spin results in no winning outcomes. A wager is required for the next spin. However, a cashout input is received and play of the game is terminated.

Spin 19: The spin example narrative provides an example of how paid spins, free spins and the application of one or more game elements to the game rules may interacts. In some cases, the game elements may be applied to paid spins and may be in effect when free spins are awarded. The game elements in effect at the time continue to be applied to the free spins and, if the free spins terminate prior to the expiration of the bonus mode, will once again be applied to the game rules during paid spins.

The recitations of a value input device for establishing a credit balance, an input device for accepting a wager input that initiates a spin, and a value output device for paying out the credit balance are integrally incorporated within the steps of the data processing method. For example, the presentation of game outcomes through the spinning and stopping of the reels is essential to the game outcome determinations, which may only be initiated by the accepted wager input. Furthermore, a value input device for establishing a credit balance, an input device for accepting a wager that initiates a spin, and a value output device for paying out the credit balance are physical, structural elements that are not shared by generic or well-known computing devices but, rather, are particular to gaming machines.

Embodiments of the present invention realize benefits in increased computer processing efficiency with minimized processing overhead, fewer rules to be evaluated, fewer player inputs to be monitored, and simpler graphical representations. With respect to the game feature triggering process, if no coin symbol appears in the array at step **310**, the game-logic circuitry foregoes any random determination of whether a game feature will be awarded. Furthermore, if any coin symbols do appear in the array, regardless of the number of pots and associated game features that may be won in parallel, only a single invocation of the RNG at step **326** is required to determine whether any game feature will actually be triggered. In contrast, in typical prior art systems with mystery bonus triggers, the game-logic circuitry makes a random determination in each and every wagering game cycle and for each and every game feature that may be won, thereby reducing processing efficiency compared to the method presented herein. Furthermore, use of a sliding range from which values are randomly selected provides multiple outcome paths with only one additional RNG call, rather than predefining an exhaustive list of possible scenarios in configuration files, memory, etc.

In this description, numerous specific details are set forth. However, it is understood that embodiments of the invention may be practiced without these specific details. For example, the displayed size of a perceived persistence/true persistence persistent element may partially represent the likelihood that its associated game feature will be triggered.

Some embodiments of the present invention comprise an innovative application of data processing steps that, when implemented by game-logic circuitry, direct a presentation assembly to present a symbol-value collection, selection, and award process that minimizes processing overhead by utilizing numbered indicia to represent credit values instead of complex, fanciful game images. In this way, these value-bearing symbols require fewer rules needed for the award process than would be necessary for calculating values of winning symbol combinations enumerated in stored paytables, as found in more complex reel-spinning routines. At the same time, embodiments of the present invention provide a straightforward, what-you-see-is-what-you-get (WYSIWYG) visual presentation that is simple to understand and, therefore, effective in generating player excitement and enthusiasm. The result is a highly flexible value-award process that can be easily adapted to any theme/brand while remaining easily understood by players.

While the examples given with respect to method **1100** portray that awarding of an additional game feature when the game is already in a bonus mode may extend the bonus mode for all currently applied game features by a certain number of spins, in some embodiments, each persistent element has its own independent bonus mode counter. Once the counter for a particular persistent element expires, that persistent element's game feature application is removed from the game rules at step **1118**. The game features associated with any other active persistent elements continue to be applied. Thus, as long as one persistent element's bonus mode counter is active, the bonus mode remains active and, when all persistent element's counters are inactive, the bonus mode is inactive. When the various game features are independently applied and removed while free spins may also be separately enabled and disabled, a rich and varied game experience results.

In other instances, well-known circuits, structures and techniques have not been shown in detail in order not to obscure the understanding of this description. Note that in this description, references to "one embodiment" or "an embodiment" mean that the feature being referred to is included in at least one embodiment of the invention. Further, separate references to "one embodiment" in this description do not necessarily refer to the same embodiment; however, neither are such embodiments mutually exclusive, unless so stated and except as will be readily apparent to those of ordinary skill in the art. Thus, the present invention can include any variety of combinations and/or integrations of the embodiments described herein. Each claim, as may be amended, constitutes an embodiment of the invention, incorporated by reference into the detailed description. Moreover, in this description, the phrase "exemplary embodiment" means that the embodiment being referred to serves as an example or illustration.

Block diagrams illustrate exemplary embodiments of the invention. Flow diagrams illustrate operations of the exemplary embodiments of the invention. The operations of the flow diagrams are described with reference to the example embodiments shown in the block diagrams. However, it should be understood that the operations of the flow diagrams could be performed by embodiments of the invention other than those discussed with reference to the block diagrams, and embodiments discussed with references to the block diagrams could perform operations different than those discussed with reference to the flow diagrams. Additionally, some embodiments may not perform all the operations shown in a flow diagram. Moreover, it should be understood that although the flow diagrams depict serial operations, certain embodiments could perform certain of those operations in parallel or in a different sequence.

Each of these embodiments and obvious variations thereof is contemplated as falling within the spirit and scope of the claimed invention, which is set forth in the following claims. Moreover, the present concepts expressly include any and all combinations and subcombinations of the preceding elements and aspects.

What is claimed is:

1. A method of operating a gaming machine, the method comprising the operations of:

accepting, via a value input device, a physical item associated with a monetary value to establish a monetary balance;

conducting, by game-logic circuitry, a game including:

presenting, by a presentation assembly, a plurality of symbol-bearing reels, an array, and a plurality of persistent elements, each persistent element having an associated game feature and an associated accumulation symbol;

spinning and stopping the plurality of reels to land symbols from the reels in the array; and

in response to the landed symbols including at least one accumulation symbol:

animating an addition of the at least one accumulation symbol to its associated persistent element; and

randomly determining whether or not to award the game feature associated with the persistent element associated with the at least one accumulation symbol; and

in response to awarding the game feature associated with the persistent element associated with the at least one accumulation symbol, implementing the game feature via the game-logic circuitry, wherein the implementing comprises applying changes associated with the game feature to game conditions currently in effect for a set of bonus spins comprising a combination of purchased spins and free spins and removing the applied changes upon completion of the bonus spins, wherein the changes associated with the game feature comprise, upon the landed symbols including a catalyst symbol, awarding values borne by one or more value-bearing symbols landed in the array; and

receiving, via at least one of one or more electronic input devices, a cash out input that initiates a payout from the monetary balance via a value output device.

2. The method of claim **1**, wherein the set of bonus spins comprises a subset of free spins preceded by a subset of purchased spins.

3. The method of claim **1**, wherein the set of bonus spins comprises a subset of purchased spins preceded by a subset of free spins.

4. The method of claim **1**, wherein the set of bonus spins comprises a predetermined number of spins.

5. The method of claim **1**, wherein the set of bonus spins comprises a randomly determined number of spins.

6. The method of claim **1**, wherein the gaming machine is regulatorily approved and primarily dedicated to playing a wagering game.

7. The method of claim **1**, wherein the animating operation includes increasing a size of the persistent element or a volume of items therein.

8. The method of claim **7**, wherein randomly determining whether or not to award the game feature associated with the persistent element is based, at least in part, on the size of the persistent element or the volume of items therein.

9. The method of claim **1**, wherein the changes associated with the game features comprise modification of the symbols borne by the plurality of reels, the modification including a wild multiplier.

10. A gaming machine comprising:

a presentation assembly;

a value input device configured to accept a physical item associated with a monetary value to establish a credit balance;

a value output device configured to dispense a payout from the credit balance in response to a cashout input; and

game-logic circuitry configured to perform the operations of:

conducting a game including:

presenting, by the presentation assembly, a plurality of symbol-bearing reels, an array, and a plurality of persistent elements, each persistent element having an associated game feature and an associated accumulation symbol;

spinning and stopping the plurality of reels to land symbols from the reels in the array; and

in response to the landed symbols including at least one accumulation symbol:

animating an addition of the at least one accumulation symbol to its associated persistent element;

randomly determining whether or not to award the game feature associated with the persistent element associated with the at least one accumulation symbol; and

in response to awarding the game feature associated with the persistent element associated with the at least one accumulation symbol, implementing the game feature via the game-logic circuitry, wherein the implementing comprises applying changes associated with the game feature to game conditions currently in effect for a set of bonus spins comprising a combination of purchased spins and free spins and removing the applied changes upon completion of the bonus spins, wherein the changes associated with the game feature comprise, upon the landed symbols including a catalyst symbol, awarding values borne by one or more value-bearing symbols landed in the array.

11. The gaming machine of claim **10**, wherein the set of bonus spins comprises a subset of free spins preceded by a subset of purchased spins.

12. The gaming machine of claim **10**, wherein the set of bonus spins comprises a subset of purchased spins preceded by a subset of free spins.

13. The gaming machine of claim **10**, wherein the set of bonus spins comprises a predetermined number of spins.

14. The gaming machine of claim **10**, wherein the set of bonus spins comprises a randomly determined number of spins.

15. The gaming machine of claim **10**, wherein the gaming machine is regulatorily approved and primarily dedicated to playing a wagering game.

16. The gaming machine of claim **10**, wherein the animating operation includes increasing a size of the persistent element or a volume of items therein.

17. The gaming machine of claim **16**, wherein randomly determining whether or not to award the game feature associated with the persistent element is based, at least in part, on the size of the persistent element or the volume of items therein.

18. The gaming machine of claim **10**, wherein the changes associated with the game features comprise modification of the symbols borne by the plurality of reels, the modification including a wild multiplier.

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